

Post-WKB Mathematical Derivations for the SU(2) Yang–Mills Kelvin–Helmholtz Program

A complete, step-by-step record of every derivation developed after `khaxn.pdf` and `wkb.pdf`

YM–KH simulation program

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Abstract

This document collects, in full step-by-step detail, all mathematical derivations produced by the SU(2) Yang–Mills Kelvin–Helmholtz (KH) program *after* the two foundational notes: `khaxn.pdf` (the linearized Yang–Mills–Wong system in the $a_x = 0$ gauge and the decoupled a_z wave equation) and `wkb.pdf` (the $\mathcal{O}(v^2)$ reduction to a Schrödinger form and the harmonic-core quantization, its eq. (33)). Results of those two notes are used *as given*; everything else is derived here from first principles. The main chain of results is: the quartic dispersion polynomial and its $k_z = 0$ chromo-Weibel limit (§2); the time-domain model as simulated, its Wong precession source and its exact energy budget including the frozen-background “battery” term (§3); the linearized six-field circular system (§4); the local dispersion relation, the tachyonic (Nielsen–Olesen–class) outer branch, and the Airy wall estimate (§5); the exact reduction of the six-field system to a single second-order ODE (§6); the precession-resonance drive, the hard growth ceiling $\gamma^3 \leq \alpha V_0^2$ and the natural units of the shear branch (§7); the quantization theory (models A and B) (§8); the confinement-set peak formula $k_z^{\text{peak}} \approx \alpha V_0(\xi_w - \ln 2) + c(\alpha/V_0)^{1/3}$ and the windowed roll-off (§9); the shear-width (quantization budget) scaling (§10); the diagnosis of the WKB quartic’s domain of validity (§11); the cold two-stream rates in code units (§12); the nonlinear cavitation threshold and the quadratic back-reaction law (§13); the warm isothermal closure (§14); and the eigenmode-seeding theory (§15). Appendices give the helicity-projection identities, the filter/sponge operators, and the numerical validation ledger.

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1 Introduction, given results, and conventions

1.1 What is taken as given

Two documents predate everything here and their results are used without re-derivation.

Given 1: the linearized Yang–Mills–Wong system (khaxn.pdf). Working in the $a_x^a = 0$ gauge with a background flow $\vec{V}_0 = V_{0z}(x)\hat{z}$, background color charge $Q_0^a = \delta^{a1}$, and color-1 background potentials $A_{0z}^1(x)$, $\phi_0^1(x)$, that note derives (its eqs. (1)–(21)) the perturbed magnetic-field components, the perturbed color charges $q^{2,3}$, and the linearized Ampère equations, and closes them (its eqs. (22)–(23)) into decoupled wave equations for the circular combinations $a_y^\pm$, $a_z^\pm$. In particular its eq. (23),

$$(-\partial_x^2 - \omega^2 + k_z^2 + \alpha^2(\phi_0^1)^2 - \alpha^2(A_{0z}^1)^2) a_z^+ = \frac{\alpha^2 V_{0z}}{\Omega + \alpha \Phi_0^1} \left(V_{0z} - \frac{k_z + \alpha A_{0z}^1}{\omega + \alpha \phi_0^1} \right) a_z^+, \quad (1.1)$$

with $\Omega = \omega - k_z V_{0z}$ and $\Phi_0^1 = V_{0z} A_{0z}^1$, is the frequency-domain ancestor of everything in §6 below; we will meet the combinations $k_z \pm \alpha A_{0z}^1$ and the beam resonance denominator $\Omega + \alpha \Phi_0^1$ again, derived independently from the time-domain system.

Given 2: the harmonic-core WKB quantization (wkb.pdf). Expanding the step-potential form of (1.1) to $\mathcal{O}(v^2)$ produces a Schrödinger form $\partial_x^2 a_z + Q_\pm(x) a_z = 0$ with

$$Q_\pm(x) = Q_0^\pm - |C_1|x^2 + C_2x^4, \quad |C_1| = \frac{\alpha^3 v^2}{2\omega^2}, \quad C_2 = \frac{\alpha^2 v^2}{4}, \quad (1.2)$$

$$Q_0^\pm = Q_{0,\text{mean}} \pm \delta, \quad Q_{0,\text{mean}} = \omega^2 - k_z^2 - \frac{\alpha^2 v k_z}{\omega^2}, \quad \delta = \frac{\alpha^2 v^2}{\omega} \left(1 - \frac{k_z^2}{\omega^2} \right). \quad (1.3)$$

The Bohr–Sommerfeld condition applied to the harmonic core, with the $\pm\delta$ jump cancelling between the two half-integrals, gives the exact quantization of the harmonic core (wkb.pdf eq. (33)):

$$\boxed{Q_{0,\text{mean}} = (2n + 1)\sqrt{|C_1|}} \quad n = 0, 1, 2, \dots \quad (1.4)$$

Everything downstream of (1.4) — the quartic polynomial, its $k_z = 0$ limit, its regime of validity, and its eventual replacement by the exact-action theory — is derived in this document.

1.2 Units, geometry, and notation

Code units are used throughout: $c = \varepsilon_0 = 1$; each beam has unperturbed density $n_0 = 1$, particle mass $m = 1$ and color-charge magnitude $|Q^1| = 1$, so that *each beam's* plasma frequency is

$$\omega_p^2 = \frac{n_0 Q^2}{m \varepsilon_0} = 1. \quad (1.5)$$

Time is measured in these plasma units (“TU”). The simulation domain is the periodic box $L_x = 6\pi$, $L_z = 2\pi$, so an integer mode index k in z has $k_z = k$ exactly. The shear layer sits at $x = L_x/2$ and all x -profiles are written in the stretched coordinate

$$\xi = \frac{x - L_x/2}{\varepsilon_s}, \quad (1.6)$$

where ε_s (called EPS in the code, typically 0.15) is the shear half-width. The two cold beams A and B counter-stream in z and carry opposite color-1 charges:

$$v_{zA} = +V_0 \tanh \xi, \quad Q_A^1 = +1; \quad v_{zB} = -V_0 \tanh \xi, \quad Q_B^1 = -1. \quad (1.7)$$

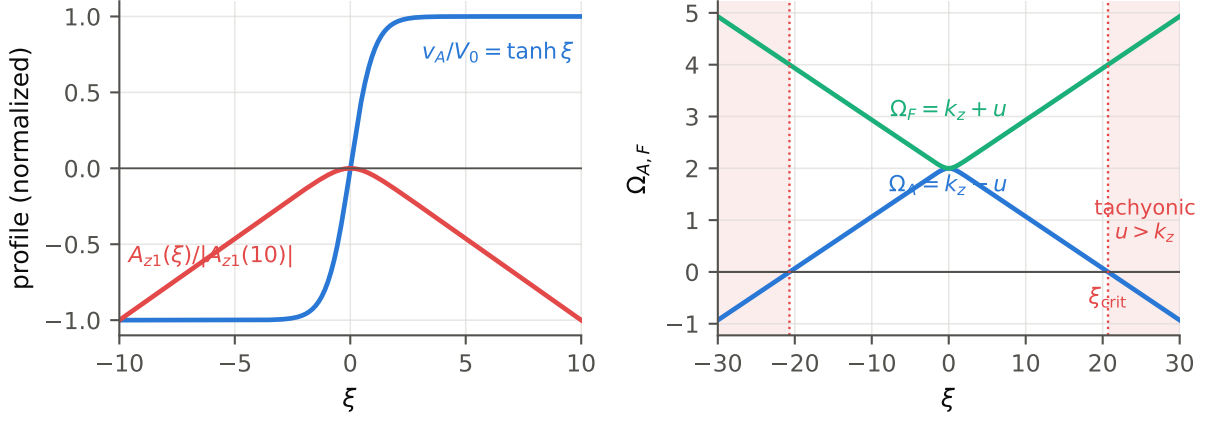


Figure 1: The production background. *Left*: counter-streaming shear $v_A = V_0 \tanh \xi$ (beam B is its mirror) and the frozen log-cosh potential A_{z1} of eq. (1.8), normalized to its value at $\xi = 10$. *Right*: the covariant wavenumbers $\Omega_A = k_z - u$ and $\Omega_F = k_z + u$ (here $\alpha = 2$, $V_0 = 0.05$, $k_z = 2$). Beyond $\xi_{\text{crit}} = k_z/(\alpha V_0) + \ln 2$ (dotted, shaded) the product $\Omega_A \Omega_F$ is negative and one circular polarization is tachyonic (§5); inside, the same shift u builds the confining well of the shear branch (§6).

The frozen color-1 background potential of the production (mode 1/6) geometry is

$$A_{z1}(\xi) = -V_0 \ln \cosh \xi, \quad \text{and we define} \quad u(\xi) \equiv -\alpha A_{z1}(\xi) = \alpha V_0 \ln \cosh \xi \geq 0. \quad (1.8)$$

For $|\xi| \gtrsim 2$, $\ln \cosh \xi = |\xi| - \ln 2 + \mathcal{O}(e^{-2|\xi|})$, so

$$u(\xi) \simeq \alpha V_0 (|\xi| - \ln 2) \quad (|\xi| \gtrsim 2). \quad (1.9)$$

$\alpha \equiv \alpha_{\text{YM}}$ is the coupling. Perturbations grow as $e^{\gamma t}$; the frequency-domain results of the two given notes map onto this via

$$\omega = i\gamma, \quad \omega^2 = -\gamma^2. \quad (1.10)$$

Two derived scales recur throughout (§7):

$$\Gamma \equiv (\alpha V_0^2)^{1/3} \text{ (rate scale / ceiling),} \quad K \equiv \left(\frac{\alpha}{V_0}\right)^{1/3} \text{ (wavenumber scale),} \quad \Gamma = V_0 K. \quad (1.11)$$

Hats denote quantities measured in these scales: $\gamma = \Gamma \hat{\gamma}$, $k_z = K \hat{k}$, etc. A window (sponge or hard cut) of radius ξ_w (in ξ -units) limits the mode region in production runs. Figure 1 shows the background profiles and the central geometric objects of the whole theory, the two shifted wavenumbers Ω_A, Ω_F derived in §4.

1.3 Reading map

Sections 2–15 are ordered logically, not chronologically: model $\rightarrow$ linearization $\rightarrow$ local theory $\rightarrow$ exact reduction $\rightarrow$ drive/ceiling $\rightarrow$ quantization $\rightarrow$ dispersion-curve consequences $\rightarrow$ scalings $\rightarrow$ nonlinear/auxiliary theory. Each section states which numerical validation anchors it; the ledger is collected in Appendix C. Indented **Physical picture** boxes carry the intuition behind each derivation. All figures are generated from the formulas derived here by `derivations/make_figs.py`; the validation figure (Fig. 8) re-runs the repository’s six-field eigensolver rather than reusing cached curves.

2 From the harmonic-core quantization to the quartic, and the $k_z = 0$ chromo-Weibel rate

2.1 Clearing eq. (33) to a quartic polynomial

Step 2.1.1. Substitute the definitions (1.2)–(1.3) into (1.4), with the background velocity scale written $v = V_0$:

$$\omega^2 - k_z^2 - \frac{\alpha^2 V_0 k_z}{\omega^2} = (2n + 1) \sqrt{\frac{\alpha^3 V_0^2}{2\omega^2}}. \quad (2.1)$$

Step 2.1.2. The square root on the right is resolved as $\sqrt{\alpha^3/2} V_0/\omega$. For real positive ω this is elementary; for the complex (growing) roots we are about to find, this choice of branch defines the analytic continuation used everywhere in the program (the continuation is fixed by requiring the $k_z = 0$ limit below to reproduce the direct cubic factorization, which it does). Define the constant

$$C \equiv (2n + 1) \sqrt{\frac{\alpha^3}{2}} V_0. \quad (2.2)$$

Then (2.1) reads

$$\omega^2 - k_z^2 - \frac{\alpha^2 V_0 k_z}{\omega^2} = \frac{C}{\omega}. \quad (2.3)$$

Step 2.1.3. Multiply through by ω^2 (legitimate: $\omega = 0$ is not a root of (2.1)) and collect all terms on one side:

$$\boxed{\omega^4 - k_z^2 \omega^2 - C \omega - \alpha^2 V_0 k_z = 0} \quad (2.4)$$

This is the *WKB quartic* (also `wkbfull.pdf` eq. (4)); the fastest-growing root $\gamma = \max \text{Im } \omega$ over its four roots is the “ γ_{WKB} ” quoted in all program tables.

Remark 2.1 (The growing root is not purely imaginary away from $k_z = 0$). An early implementation assumed $\text{Re } \omega = 0$ and bisected on real γ . Substituting $\omega = i\gamma$ into (2.4) gives $\gamma^4 + k_z^2 \gamma^2 - \alpha^2 V_0 k_z = iC\gamma$, whose left side is real and right side imaginary — so a purely imaginary root exists only where $C\gamma$ can vanish, i.e. at $k_z = 0$ (where the real part balances differently, see below) or in the trivial limit $C \rightarrow 0$. Numerically the purely-imaginary assumption erred by up to $\sim 50\%$ at $k_z = 0.25$ and $\sim 10\%$ at $k_z = 1$, shrinking with k_z ; the full complex quartic is solved everywhere in current tooling.

2.2 The $k_z = 0$ limit: chromo-Weibel growth rate

Step 2.2.1. Set $k_z = 0$ in (2.4). Both the $k_z^2 \omega^2$ term and the constant term $\alpha^2 V_0 k_z$ vanish:

$$\omega^4 - C\omega = 0 \quad \Longleftrightarrow \quad \omega(\omega^3 - C) = 0. \quad (2.5)$$

Step 2.2.2. Discard the root $\omega = 0$ (no dynamics). The cubic factor gives the three cube roots of the positive real number C :

$$\omega \in \left\{ C^{1/3}, C^{1/3} e^{2\pi i/3}, C^{1/3} e^{-2\pi i/3} \right\}. \quad (2.6)$$

Step 2.2.3. The growing root is the one with positive imaginary part, $\omega = C^{1/3} e^{2\pi i/3}$, with

$$\gamma = \text{Im } \omega = C^{1/3} \sin \frac{2\pi}{3} = \frac{\sqrt{3}}{2} C^{1/3}. \quad (2.7)$$

Step 2.2.4. Insert C from (2.2) for the fundamental $n = 0$:

$$\boxed{\gamma_{\text{Weibel}}(k_z=0) = \left(\sqrt{\frac{\alpha^3}{2}} V_0 \right)^{1/3} \frac{\sqrt{3}}{2} = \left(\sqrt{\frac{\alpha^3}{2}} V_0 \right)^{1/3} \sin 60^\circ} \quad (2.8)$$

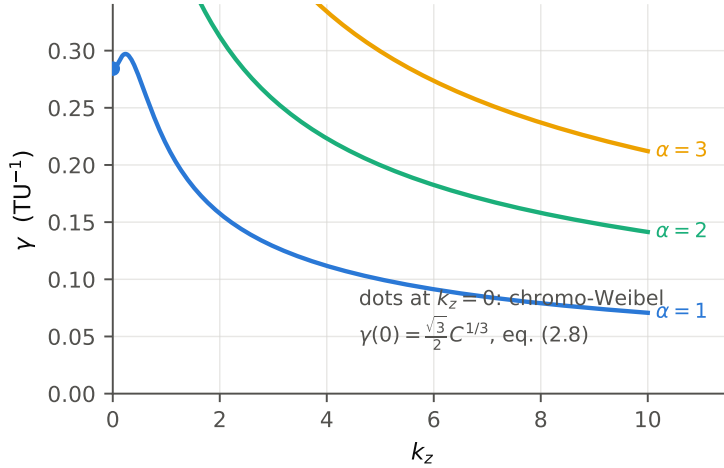


Figure 2: The WKB quartic (2.4) at $V_0 = 0.05$ for $\alpha = 1, 2, 3$ ($n = 0$), with the exact $k_z = 0$ chromo-Weibel roots (2.8) (dots). The $k_z = 0$ anchor is validated to 0.5% in simulation; the finite- k_z behaviour of the quartic is its harmonic-core extrapolation, superseded by the exact-action theory — see the diagnosis in §11 and the direct comparison in Fig. 8.

Remark 2.2 (Physical content and validation). At $k_z = 0$ the unstable channel is a color-mixing filamentation (“chromo-Weibel”) mode: the background A_{z1} mixes the color-2 and color-3 electromagnetic fields through the non-Abelian Ampère term $-\alpha(A_z \times B_y)^a$ and Faraday term $+\alpha(A_z \times E_x)^a$ (§3), producing z -uniform exponentially growing color-2/3 fields. Direct simulation reproduced (2.8) to 0.5%, the strongest single anchor of the quartic. This mode is deliberately suppressed in production KH campaigns (`suppress_kz0=1` plus hyperdiffusion, Appendix B); at the reference point of the self-consistency study (§13.2), $\alpha = 1$, $V_0 = 0.05$, formula (2.8) gives $\gamma = 0.284$, exactly the measured irreducible $k_z = 0$ growth there. Figure 2 shows the quartic’s dispersion curves with the $k_z = 0$ anchors.

Physical picture. At $k_z = 0$ nothing varies along the flow, so this is not a bunching instability but *filamentation*: the frozen A_{z1} rotates the color-2/3 fields into each other while the counter-streaming color currents feed them, and the displaced currents reinforce the seed field. The cube root in (2.8) is the signature of the loop: closing the feedback ($E \rightarrow A$ by one time integration, $A \rightarrow Q$ by a second, $Q \rightarrow J \rightarrow E$ by a third) gives a loop gain $\propto C/\gamma^3$; marginal self-amplification, gain of order one, fixes $\gamma \sim C^{1/3}$, and the exact root geometry supplies the $\sin 60^\circ$.

3 The time-domain model as posed

The nonlinear system integrated by the CUDA code is the time-domain counterpart of the linearized system of `khaxn.pdf`. It is stated here as the model definition; the derivations in this section are (i) the Wong-equation origin of the precession source, (ii) the invariant $B_y^a = -\partial_x A_z^a$ of the potential/Faraday pair, and (iii) the exact energy budget, which identifies the frozen background as an external battery.

3.1 Model equations

Fields: for each color $a \in \{1, 2, 3\}$, the 2D (x, z) fields $E_x^a, E_z^a, B_y^a, A_z^a$; for each beam $s \in \{A, B\}$, the fluid fields n_s, p_{xs}, p_{zs} (momentum densities; $v_{is} = p_{is}/n_s$) and the color-charge densities Q_s^a . Only A_z^a is carried as an explicit potential ($\partial_y \equiv 0$; no y -components of E or of the fluid velocity are excited by the dynamics below). The gauge choice this reflects, and the precise (non-)status of A_x^a and A_y^a , is examined in §3.2 immediately below: in short, $A_0^a \equiv 0$ is the one legitimate

gauge condition spent here, while the absence of A_y^a and A_x^a are two separate statements of a different kind, not a second gauge-fixing.

Fluid sector (each beam; FCT advection):

$$\partial_t n + \nabla \cdot (n \vec{v}) = 0, \quad (3.1)$$

$$\partial_t p_x + \nabla \cdot (\vec{v} p_x) = F_x = -n \sum_a Q^a (E_x^a + v_z B_y^a), \quad (3.2)$$

$$\partial_t p_z + \nabla \cdot (\vec{v} p_z) = F_z = -n \sum_a Q^a (E_z^a - v_x B_y^a), \quad (3.3)$$

$$\partial_t Q^a + \nabla \cdot (\vec{v} Q^a) = \alpha v_z (\vec{Q} \times \vec{A}_z)^a, \quad (3.4)$$

where $(\vec{Q} \times \vec{A}_z)^a = \epsilon^{abc} Q^b A_z^c$ is the SU(2) adjoint cross product. (In (3.4) the Q^a are advected as densities; combined with (3.1) this is equivalent to the advective form $dQ^a/dt = \alpha v_z (\vec{Q} \times \vec{A}_z)^a$ along fluid trajectories, which is what Wong's equation supplies, §3.3.)

Field sector (leapfrog on a Yee stagger):

$$\partial_t E_x^a = -\partial_z B_y^a - \alpha (\vec{A}_z \times \vec{B}_y)^a - J_x^a, \quad (3.5)$$

$$\partial_t E_z^a = +\partial_x B_y^a - J_z^a, \quad (3.6)$$

$$\partial_t B_y^a = \partial_x E_z^a - \partial_z E_x^a + \alpha (\vec{A}_z \times \vec{E}_x)^a, \quad (3.7)$$

$$\partial_t A_z^a = -E_z^a \quad (A_z^1 \text{ held fixed when } \texttt{freeze_az1}=1), \quad (3.8)$$

$$J_i^a = -n_A Q_A^a v_{iA} - n_B Q_B^a v_{iB}, \quad i \in \{x, z\}. \quad (3.9)$$

Equation (3.6) carries no non-Abelian term: the candidate commutator $\alpha \epsilon^{abc} A_z^b F^{zz,c}$ vanishes identically because only the z -component of the potential is kept. The sign conventions (charge enters the current and the force with an overall minus) are the model's own; their only invariant consequences are the products current $\times$ field and force $\times$ velocity, and those are fixed by the energy budget below.

3.2 Gauge counting: what the field content assumes

The field content just declared silently makes a claim worth checking explicitly: that dropping A_x^a and A_y^a while keeping only A_z^a is licensed by gauge freedom. It is not, quite — the SU(2) gauge group supplies exactly enough freedom to eliminate *one* full spacetime-component of A_μ^a , not two, and the argument for why the model is nonetheless consistent has two genuinely different parts.

Step 3.2.1 (The counting). An infinitesimal SU(2) gauge transformation is parametrized by $\theta^a(t, x, z)$, $a = 1, 2, 3$: three arbitrary functions of the three coordinates the fields depend on here ($\partial_y \equiv 0$ removes y from the list), transforming $\delta A_\mu^a = -(D_\mu \theta)^a$. A condition such as “ $A_x^a(t, x, z) = 0$ for all (t, x, z) ” is three equations (one per color) holding at every point of a three-dimensional domain — exactly as many conditions as gauge functions available. Generically this exhausts the freedom (up to a residual gauge transformation independent of x , the integration freedom left when solving $\partial_x \theta^a = (\dots)$ pointwise in x). The general statement:

one gauge choice $\implies$ one full spacetime component of A_μ^a eliminated — not two.

(3.10)

Step 3.2.2 (What `khaxn.pdf` spends its one choice on). Its stated gauge is $a_x^a = 0$, and nothing else. In particular:

- $\phi \equiv A_0$ is *not* set to zero. It survives as a genuine background field ($E_{0x}^1 = -\partial_x \phi_0^1$, its §1) and perturbation field, eliminated only via the Gauss-law *constraint* — a fundamentally different move from a gauge choice — in its own “Substitution and Simplification” section, solving ϕ^+ in terms of a_z^+ .

- a_y^a is not set to zero either: it is explicitly retained and sources b_x (its eq. 1). Its own “Final equations” §2 shows $a_y^\pm$ (eq. 22) and $a_z^\pm$ with ϕ eliminated (eq. 23) form two separate equations, not interacting with each other at this order.

The surviving content after `khaxn.pdf`’s one gauge choice is therefore (ϕ, a_y, a_z) , not a_z alone.

Step 3.2.3 (What the simulated model spends its one choice on). The potential equation (3.8), $\partial_t A_z^a = -E_z^a$, is the exact temporal-gauge relation $E_z = -\partial_t A_z$ with no $-\partial_z A_0$ correction term. This identifies the model’s one gauge condition as

$$A_0^a \equiv 0 \quad (\text{temporal gauge}), \quad (3.11)$$

a legitimate, complete choice — and a *different* one from `khaxn.pdf`’s axial choice, not an additional condition stacked on top of it.

Step 3.2.4 (Why A_y^a and A_x^a can then be dropped without spending more freedom). Two separate arguments, of two different kinds:

- A_y^a is a sector choice, not a gauge choice. The $a_y^\pm$ branch (`khaxn.pdf` eq. 22) does not source, or get sourced by, the a_z branch; the perturbation studied here (By2-type seeding, §15) never populates it. Omitting it selects *which already-decoupled physical sector to simulate* — licensed by the decoupling `khaxn.pdf` itself exhibits, not by spending extra gauge freedom. (Restated as the scope remark of §4.4.)
- A_x^a is wired to nothing. Unlike A_y^a , this field does not merely go unsourced by the chosen initial data — it never appears, at all, in any covariant or commutator term of eqs. (3.2)–(3.9): every non-Abelian correction in the model (Ampère- x , Faraday, precession) references $\vec{A}_z$ only. A hypothetical A_x^a generated by some other mechanism would still have zero effect on any other field’s equation of motion; it is a complete spectator, consistently left untracked regardless of its own value.

Physical picture. One knob, one turn. Gauge freedom is a redundancy in the description, not in the physics: the count of gauge functions must match the count of unphysical directions removed. `khaxn.pdf` turns the “ A_x ” knob and leaves ϕ, a_y as real content — one eliminated by a constraint, the other by non-interaction with the sector of interest. This model turns a *different* knob, “ A_0 ” — equally legitimate, equally a single turn — and then drops A_y for a decoupling reason and A_x because it is wired to nothing. Two different single choices of gauge are not the same thing as one gauge choice spent twice.

Remark 3.1 (Consistency check: E_x is not spurious content). If the model had silently fixed both $A_x = 0$ and $A_0 = 0$, its would-be substitute for ϕ — namely E_x — ought to carry independent new physics absent from `khaxn.pdf`’s construction. It does not: the linearized Ampère- x equation derived in §4, eq. (4.9) ($\gamma e_x = -i\Omega_{Ab}$), shows e_x is purely and instantaneously slaved to b — a first-order algebraic relation, not an independently propagating mode. This is exactly the behavior a properly constraint-eliminated ϕ must have: E_x here plays ϕ ’s role in temporal-gauge dress, rather than adding unphysical content the given theory does not have.

3.3 Derivation of the precession source from Wong’s equations

Step 3.3.1. Wong’s equation for the color charge of a classical particle moving on a worldline $x^\mu(\tau)$ in a background potential A_μ^b is

$$\frac{dQ^a}{d\tau} + g \epsilon^{abc} \frac{dx^\mu}{d\tau} A_\mu^b Q^c = 0. \quad (3.12)$$

Step 3.3.2. Non-relativistic limit ($\tau \rightarrow t$, $dx^\mu/d\tau \rightarrow (1, \vec{v})$), temporal gauge $A_0^b = 0$, and only $A_z^b \neq 0$:

$$\frac{dQ^a}{dt} = -g \epsilon^{abc} v_z A_z^b Q^c. \quad (3.13)$$

Step 3.3.3. Rename $g \rightarrow \alpha$ and swap the summed indices $b \leftrightarrow c$ using antisymmetry, $\epsilon^{abc} A^b Q^c = -\epsilon^{abc} Q^b A^c$:

$$\frac{dQ^a}{dt} = +\alpha v_z \epsilon^{abc} Q^b A_z^c = \alpha v_z (\vec{Q} \times \vec{A}_z)^a, \quad (3.14)$$

which is exactly the source in (3.4). Because $|\vec{Q}|$ is conserved by a cross-product rotation, $Q_{A,B}^1 = \pm 1$ remains a unit adjoint vector until advection mixes fluid elements.

3.4 The invariant $B_y^a = -\partial_x A_z^a$ of the potential pair

From (3.8), $\partial_t(-\partial_x A_z^a) = \partial_x E_z^a$. Comparing with (3.7), the difference field $W^a \equiv B_y^a + \partial_x A_z^a$ obeys

$$\partial_t W^a = -\partial_z E_x^a + \alpha (\vec{A}_z \times \vec{E}_x)^a. \quad (3.15)$$

Thus $B_y = -\partial_x A_z$ (the magnetostatic relation, matching `khaxn.pdf` eqs. (4)–(6), $b_y^a = -\partial_x a_z^a$) is preserved exactly for z -uniform, color-aligned configurations (e.g. the color-1 background itself, for which the right side of (3.15) vanishes), but *not* for the color-2/3 wave sector, where B_y carries the additional dynamics injected through E_x . This is why the code treats B_y^a as an independent evolved field, and why a current-consistent B_{y1} initialization (`init.by1_eq`) is needed when the color-1 sector is live (§13.2).

3.5 Exact energy budget: conservative core plus the frozen-background battery

The monitored energy is

$$E(t) = \int dx dz \left[\frac{1}{2} \sum_a ((E_x^a)^2 + (E_z^a)^2 + (B_y^a)^2) + \sum_{s=A,B} \frac{p_{xs}^2 + p_{zs}^2}{2n_s} \right]. \quad (3.16)$$

We differentiate term by term using (3.1)–(3.9), at the continuum level and over the periodic box (so all pure divergences integrate to zero).

Step 3.5.1 (Setting up the calculation). Differentiate (3.16) under the integral sign. The field part is immediate from the product rule,

$$\frac{d}{dt} \left[\frac{1}{2} \sum_a ((E_x^a)^2 + (E_z^a)^2 + (B_y^a)^2) \right] = \sum_a \left[E_x^a \partial_t E_x^a + E_z^a \partial_t E_z^a + B_y^a \partial_t B_y^a \right], \quad (3.17)$$

and each $\partial_t(\cdot)$ is then replaced by the *full* right-hand side of (3.5)–(3.7) — curl term, current term, and non-Abelian commutator term alike; nothing is dropped at this stage, and the three steps below simply sort the resulting terms into three physically distinct piles.

The fluid part needs one extra piece of algebra because n_s also depends on t . Write $K_i \equiv p_i^2/(2n)$ for either momentum component of either beam (beam label suppressed) and use the chain rule together with (3.1)–(3.3):

$$\partial_t K_i = \frac{p_i}{n} \partial_t p_i - \frac{p_i^2}{2n^2} \partial_t n = v_i [-\nabla \cdot (\vec{v} p_i) + F_i] - \frac{v_i^2}{2} [-\nabla \cdot (n \vec{v})]. \quad (3.18)$$

Expand both divergences with the product rule, using $p_i = n v_i$: $\nabla \cdot (\vec{v} p_i) = v_i \nabla \cdot (n \vec{v}) + n \vec{v} \cdot \nabla v_i$. Substituting and collecting the two $\nabla \cdot (n \vec{v})$ pieces (coefficients $-v_i^2$ and $+v_i^2/2$, net $-v_i^2/2$):

$$\partial_t K_i = -\frac{1}{2} v_i^2 \nabla \cdot (n \vec{v}) - n v_i (\vec{v} \cdot \nabla v_i) + v_i F_i. \quad (3.19)$$

Compare with $\nabla \cdot (\vec{v} K_i) = \nabla \cdot (\vec{v} n v_i^2/2) = \frac{1}{2} v_i^2 \nabla \cdot (n \vec{v}) + n v_i (\vec{v} \cdot \nabla v_i)$ (same product rule, run on K_i directly): the right side above is exactly $-\nabla \cdot (\vec{v} K_i)$ plus the leftover $v_i F_i$. Hence the fluid kinetic-energy identity used below,

$$\partial_t K_i = -\nabla \cdot (\vec{v} K_i) + v_i F_i, \quad (3.20)$$

holding term by term for $i \in \{x, z\}$ and each beam: transport is conservative (a pure divergence, integrates to zero over the periodic box), and the only source is $v_i F_i$ — structurally identical to the field energy, where transport is the Poynting divergence and the only source is the work $\vec{J} \cdot \vec{E}$ plus the non-Abelian terms.

Step 3.5.2 (Transport terms). The Maxwell curl terms combine into divergences:

$$E_x^a(-\partial_z B_y^a) + B_y^a(-\partial_z E_x^a) = -\partial_z(E_x^a B_y^a), \quad E_z^a(\partial_x B_y^a) + B_y^a(\partial_x E_z^a) = \partial_x(E_z^a B_y^a), \quad (3.21)$$

i.e. a Poynting flux; both integrate to zero. Likewise, for smooth flow the FCT advection terms transport $\frac{1}{2}p^2/n$ conservatively: using (3.1)–(3.3) without sources, $\partial_t(p_i^2/2n) + \nabla \cdot (\vec{v} p_i^2/2n) = 0$.

Step 3.5.3 (Current work cancels against the Lorentz work). The field sector loses $\sum_a (J_x^a E_x^a + J_z^a E_z^a) = \vec{J}^a \cdot \vec{E}^a$; with (3.9) this equals $-\sum_s n_s \sum_a Q_s^a \vec{v}_s \cdot \vec{E}^a$. The fluid sector gains $\sum_s \vec{v}_s \cdot \vec{F}_s$ with $\vec{F}_s$ from (3.2)–(3.3); the magnetic parts cancel within the dot product,

$$v_x(v_z B_y^a) + v_z(-v_x B_y^a) = 0, \quad (3.22)$$

leaving $\sum_s \vec{v}_s \cdot \vec{F}_s = -\sum_s n_s \sum_a Q_s^a \vec{v}_s \cdot \vec{E}^a$. The two contributions are equal and opposite: energy flows between fields and fluid with no loss. (This is the invariant content of the model's charge-sign convention: the same minus sign appears in (3.9) and in (3.2)–(3.3), so all work terms pair correctly.)

Step 3.5.4 (Non-Abelian cross terms: the battery). Collect what is left after the previous two steps. Substitute the full right-hand sides of (3.5) and (3.7) into the field part of dE/dt set up above, and the full (3.2)–(3.3) into the fluid part via (3.20). Every term is now accounted for except two: the commutator piece of Ampère- x , multiplied by E_x^a , and the commutator piece of Faraday, multiplied by B_y^a (the curl pieces became the Poynting flux of the transport step and integrate to zero; the current pieces cancelled against the Lorentz-force work of the previous step; the fluid transport pieces of (3.20) also integrate to zero, same step). What remains is

$$\left. \frac{dE}{dt} \right|_{\text{NA}} = \int \left[\underbrace{E_x^a \cdot (-\alpha(\vec{A}_z \times \vec{B}_y)^a)}_{\text{leftover from Ampère-}x} + \underbrace{B_y^a \cdot (+\alpha(\vec{A}_z \times \vec{E}_x)^a)}_{\text{leftover from Faraday}} \right], \quad (3.23)$$

i.e., writing out the two commutators (sum over the repeated color index a understood in each product),

$$\left. \frac{dE}{dt} \right|_{\text{NA}} = \int \left[-\alpha \sum_a E_x^a (\vec{A}_z \times \vec{B}_y)^a + \alpha \sum_a B_y^a (\vec{A}_z \times \vec{E}_x)^a \right]. \quad (3.24)$$

Step 3.5.5 (An epsilon-tensor identity). Write both terms as scalar triple products in color space, $[\vec{X}, \vec{Y}, \vec{Z}] \equiv \epsilon^{abc} X^a Y^b Z^c$ (sum over $a, b, c \in \{1, 2, 3\}$; the first vector supplies the free index that is also the ϵ tensor's first slot), so $\sum_a E_x^a (\vec{A}_z \times \vec{B}_y)^a = [\vec{E}_x, \vec{A}_z, \vec{B}_y]$ and $\sum_a B_y^a (\vec{A}_z \times \vec{E}_x)^a = [\vec{B}_y, \vec{A}_z, \vec{E}_x]$. We need the relation between these two. Start from the second and relabel the three *dummy* summation indices $a \rightarrow c, b \rightarrow b, c \rightarrow a$ (legal: summed indices carry no meaning of their own):

$$[\vec{B}_y, \vec{A}_z, \vec{E}_x] = \epsilon^{abc} B_y^a A_z^b E_x^c = \epsilon^{cba} B_y^c A_z^b E_x^a = \epsilon^{cba} E_x^a A_z^b B_y^c. \quad (3.25)$$

The index string ϵ^{cba} differs from ϵ^{abc} only by swapping the 1st and 3rd slots ($a \leftrightarrow c$) — a single transposition, which flips the sign of the totally antisymmetric symbol: $\epsilon^{cba} = -\epsilon^{abc}$ (spot check: $\epsilon^{213} = -\epsilon^{123} = -1$, consistent with the definition $\epsilon^{123} = +1$ and total antisymmetry). Hence

$$[\vec{B}_y, \vec{A}_z, \vec{E}_x] = -\epsilon^{abc} E_x^a A_z^b B_y^c = -[\vec{E}_x, \vec{A}_z, \vec{B}_y]. \quad (3.26)$$

In words: swapping the outer two vectors of a triple product costs a minus sign, with the middle vector as a spectator. This is the general rule behind every “ $[\dots] = -[\dots]$ ” step used in the program.

Step 3.5.6 (Assembling the battery term). Insert (3.26) into the result of the first step of this sequence:

$$\begin{aligned} \left. \frac{dE}{dt} \right|_{\text{NA}} &= -\alpha \int [\vec{E}_x, \vec{A}_z, \vec{B}_y] + \alpha \int [\vec{B}_y, \vec{A}_z, \vec{E}_x] \\ &= -\alpha \int [\vec{E}_x, \vec{A}_z, \vec{B}_y] - \alpha \int [\vec{E}_x, \vec{A}_z, \vec{B}_y] = -2\alpha \int [\vec{E}_x, \vec{A}_z, \vec{B}_y]. \end{aligned} \quad (3.27)$$

Finally translate back to an ordinary dot-and-cross product: $[\vec{E}_x, \vec{A}_z, \vec{B}_y] = \vec{A}_z \cdot (\vec{B}_y \times \vec{E}_x)$. Proof: in components, $(\vec{B}_y \times \vec{E}_x)^b = \epsilon^{bca} B_y^c E_x^a$ (the standard formula, indices cycled), so

$$\vec{A}_z \cdot (\vec{B}_y \times \vec{E}_x) = A_z^b \epsilon^{bca} B_y^c E_x^a = \epsilon^{bca} E_x^a A_z^b B_y^c. \quad (3.28)$$

The index string bca is a *cyclic* permutation of abc ($a \rightarrow b \rightarrow c \rightarrow a$), and cyclic permutations of ϵ are always even (no sign change: two transpositions), so $\epsilon^{bca} = \epsilon^{abc}$, and the two expressions agree term by term: $\vec{A}_z \cdot (\vec{B}_y \times \vec{E}_x) = \epsilon^{abc} E_x^a A_z^b B_y^c = [\vec{E}_x, \vec{A}_z, \vec{B}_y]$. Therefore

$$\left. \frac{dE}{dt} \right|_{\text{NA}} = -2\alpha \int \vec{A}_z \cdot (\vec{B}_y \times \vec{E}_x). \quad (3.29)$$

This does *not* vanish in general: the model's quadratic energy (3.16) is not the full Yang–Mills energy (which hides the interaction energy inside the field-strength commutators), and (3.29) is precisely the exchange with the potential sector.

Step 3.5.7 (Evaluation on the frozen background). Unlike $\vec{A}_z$, whose color-2/3 components really are pure wave perturbation (absent from the background, §4), $\vec{B}_y$ and $\vec{E}_x$ each already have a nonzero color-1 *background* piece in general — this is not automatic and must be checked separately for each, since neither is protected by the same “wave-only” argument that lets $\vec{A}_z \approx (A_{z1}, a_2, a_3)$ with only a_2, a_3 small.

- E_x^1 : in steady state, its source in Ampère- x (3.5) is $-\partial_z B_y^1 - \alpha(\vec{A}_z \times \vec{B}_y)^1 - J_x^1$. The background is z -independent (kills the curl term); the commutator $(\vec{A}_z \times \vec{B}_y)^1 = A_z^2 B_y^3 - A_z^3 B_y^2$ needs a background $A_z^{2,3}$, which is zero; and $J_x^1 = 0$ since neither beam carries background v_x (1.7). So E_x^1 has no background source at all, and (matching the code's initialization) $E_x^1 \equiv 0$ in the background is exact, not an approximation.
- B_y^1 : by contrast, the invariant of §3.4, $B_y^a = -\partial_x A_z^a$, applies to the z -uniform color-1 background exactly, so $B_y^1 = -\partial_x A_{z1}(\xi) \neq 0$ whenever A_{z1} has x -structure — which it always does (it is the shear-coupling background). This is a genuine, nonzero background field, consistent with the code's own gauge-consistent initialization of By1.

Expand the full dot product $\vec{A}_z \cdot (\vec{B}_y \times \vec{E}_x) = \sum_b A_z^b (\vec{B}_y \times \vec{E}_x)^b$ term by term in b , using $\vec{A}_z = (A_{z1}, a_2, a_3)$, $\vec{B}_y = (B_y^1, B_y^2, B_y^3)$, $\vec{E}_x = (0, E_x^2, E_x^3)$ (the just-established $E_x^1 = 0$): the $b=1$ term gives $A_{z1}(B_y^2 E_x^3 - B_y^3 E_x^2)$ exactly as before; the $b=2, 3$ terms, worked out the same way as (4.4)–(4.5) but now with $E_x^1 = 0$, collapse to $B_y^1(a_3 E_x^2 - a_2 E_x^3) = B_y^1 \text{Im}(a e_x^*)$ (circular pairs $a = a_2 + ia_3$, $e_x = E_x^2 + iE_x^3$; check by direct expansion: $a e_x^* = (a_2 + ia_3)(E_x^2 - iE_x^3)$ has imaginary part $a_3 E_x^2 - a_2 E_x^3$). So, exactly,

$$\vec{A}_z \cdot (\vec{B}_y \times \vec{E}_x) = A_{z1} (B_y^2 E_x^3 - B_y^3 E_x^2) + B_y^1 \text{Im}(a e_x^*). \quad (3.30)$$

The second term is the same order in the wave amplitude as the first: both are one factor of background times two factors of wave. It is *not* generally negligible on formal grounds. In the production geometry (mode 5, §4) it happens to be small, but for a reason specific to that background's geometry, not a perturbative smallness. $A_{z1} = -V_0 \cos(k_x x)$ is engineered to peak, flat to second order, exactly at the shear layer where the KH eigenfunction is supported. Its

derivative $B_y^1 = -V_0 k_x \sin(k_x x)$ vanishes there and grows only linearly with distance from the layer, over that same support. So the two factors in the second term are small together, in the one region that matters, even though B_y^1 's *global* maximum (far from the layer, where the eigenfunction has already decayed) is not small. This is the same omission already flagged, and bounded empirically at the few-percent level against the full simulation, in the six-field system's scope remark (§4.4, item 2); it is not re-derived here, and the rest of this step keeps only the leading (first) term, so every result downstream of (3.35) inherits this few-percent-level caveat.

Substituting the leading term into (3.29):

$$\left. \frac{dE}{dt} \right|_{\text{NA}} = -2\alpha \int A_{z1} (B_y^2 E_x^3 - B_y^3 E_x^2). \quad (3.31)$$

Now introduce the same complex “circular” pairs used throughout §4, $b = B_y^2 + iB_y^3$ and $e_x = E_x^2 + iE_x^3$ — at this point purely a notational bundling of two real numbers into one complex one, nothing physical yet. Multiply out

$$b^* e_x = (B_y^2 - iB_y^3)(E_x^2 + iE_x^3) = \underbrace{(B_y^2 E_x^2 + B_y^3 E_x^3)}_{\text{Re}(b^* e_x)} + i \underbrace{(B_y^2 E_x^3 - B_y^3 E_x^2)}_{\text{Im}(b^* e_x)}, \quad (3.32)$$

so the bracket above is exactly $\text{Im}(b^* e_x)$:

$$\left. \frac{dE}{dt} \right|_{\text{NA}} = -2\alpha \int A_{z1} \text{Im}(b^* e_x). \quad (3.33)$$

Now use the linearized Ampère- x relation derived independently in §4, $\gamma e_x = -i\Omega_A b$ (eq. (4.9) below) — legitimate here because it is simply the statement that e_x and b are not independent once the equations of motion hold. Solve for $e_x = -i\Omega_A b/\gamma$ (real γ , Ω_A for a mode on the real- γ branch) and substitute:

$$\begin{aligned} b^* e_x &= b^* \left(\frac{-i\Omega_A b}{\gamma} \right) = \frac{-i\Omega_A}{\gamma} b^* b \\ &= \frac{-i\Omega_A}{\gamma} |b|^2, \end{aligned} \quad (3.34)$$

using $b^* b = |b|^2$ (real, non-negative). A number of the form $-i \times (\text{real})$ has $\text{Im} = -(\text{that real number})$, so $\text{Im}(b^* e_x) = -(\Omega_A/\gamma)|b|^2$. Substituting back into (3.33):

$$\left. \frac{dE}{dt} \right|_{\text{NA}} = -2\alpha \int A_{z1} \left(-\frac{\Omega_A}{\gamma} |b|^2 \right) = \frac{2\alpha}{\gamma} \int A_{z1} \Omega_A |b|^2. \quad (3.35)$$

In the tachyonic outer region (§5) $\Omega_A = k_z - u < 0$ while $A_{z1} < 0$: the integrand is positive — the frozen background *feeds* the wave, and with `freeze_az1=1` the battery is bottomless. In the shear-branch well $\Omega_A > 0$ and the same term is an energy drain; the shear branch instead grows at the expense of the beams' kinetic energy through the (internally cancelling) current-work channel. Equation (3.35) is the energetic statement of the freeze artifact discussed in §5.2 and measured in §13.2.

Physical picture. A frozen field is an infinite battery. Holding A_{z1} fixed is the field-theory analogue of an external magnet held in place by an unmodelled agent: any process that extracts energy from it can run forever, and (3.29) is the wire through which the charge flows. The tachyonic branch of §5 is precisely the discharge of this battery into charged waves — the 2D analogue of the decay of a constant chromomagnetic background (Nielsen–Olesen). The shear branch draws on a different account, the beams' bulk kinetic energy, which is a legitimate, self-consistently modelled reservoir — which is why windowing away the battery-powered branch while keeping the flow-powered one is a physically meaningful excision and not a fudge.

Remark 3.2 (Covariant Gauss constraint). The monitored (not enforced) constraint is the covariant Gauss law $\partial_x E_x^a + \partial_z E_z^a + \alpha \epsilon^{abc} A_z^b E_z^c = \rho^a$ with $\rho^a = -\sum_s n_s Q_s^a$, checked numerically by `gauss_check.py`; the leapfrog scheme preserves it to discretization accuracy in the same way the standard Yee scheme preserves the Abelian constraint.

4 Linearization: the six-field circular system

This section derives, in full detail, the linear eigenvalue system implemented by the 1D solver `ym_eigenmode.py` and reduced by all later theory. It was validated against the 2D simulation at the 1–4% level across the production grid.

4.1 Background and perturbation ansatz

Background (mode 1/6 geometry): beams (1.7), frozen potential (1.8), all other fields zero. Write $v(\xi) \equiv V_0 \tanh \xi$ so $v_{zA} = v$, $v_{zB} = -v$. Perturbations of the color-2/3 sector are combined into complex “circular” pairs

$$b = B_y^2 + iB_y^3, \quad e_x = E_x^2 + iE_x^3, \quad e_z = E_z^2 + iE_z^3, \quad a = A_z^2 + iA_z^3, \quad q_{A,B} = Q_{A,B}^2 + iQ_{A,B}^3, \quad (4.1)$$

each with the ansatz $f(x, z, t) = \hat{f}(x) e^{ik_z z + \gamma t}$, i.e.

$$\partial_t \rightarrow \gamma, \quad \partial_z \rightarrow ik_z. \quad (4.2)$$

Why circular pairs work: every non-Abelian term in (3.5)–(3.8) and (3.4) is a cross product with the color-1 background, and this always reduces to a single phase multiplication. Recall the general cross-product component formula $(\vec{Y} \times \vec{X})^i = \epsilon^{ijk} Y^j X^k$, i.e. $(\vec{Y} \times \vec{X})^1 = Y^2 X^3 - Y^3 X^2$, $(\vec{Y} \times \vec{X})^2 = Y^3 X^1 - Y^1 X^3$, $(\vec{Y} \times \vec{X})^3 = Y^1 X^2 - Y^2 X^1$. Apply this with $\vec{Y} = \vec{A}_z = (A_{z1}, 0, 0)$ (only $Y^1 = A_{z1}$ nonzero) and a generic color vector $\vec{X} = (X^1, X^2, X^3)$ whose own color-1 component X^1 is either absent (the perturbed color-2/3 fields have no color-1 part by definition) or decoupled at this order (see the parenthetical below):

$$(\vec{A}_z \times \vec{X})^2 = Y^3 X^1 - Y^1 X^3 = 0 - A_{z1} X^3 = -A_{z1} X^3, \quad (4.3)$$

$$(\vec{A}_z \times \vec{X})^3 = Y^1 X^2 - Y^2 X^1 = A_{z1} X^2 - 0 = +A_{z1} X^2. \quad (4.4)$$

Combine these as (2) + i(3) and factor:

$$(\vec{A}_z \times \vec{X})^2 + i(\vec{A}_z \times \vec{X})^3 = -A_{z1} X^3 + iA_{z1} X^2 = iA_{z1} (X^2 + iX^3) = iA_{z1} X^c, \quad (4.5)$$

where $X^c \equiv X^2 + iX^3$ and the middle equality is checked by expanding the right side back out: $iA_{z1} (X^2 + iX^3) = iA_{z1} X^2 - A_{z1} X^3$, which matches term by term. (Contributions with the perturbed a against background color-2/3 fields vanish because the latter are zero; the color-1 wave fields B_y^1, E_x^1 couple only at second order or through backgrounds excluded from this model — see the scope remark in §4.4.) A background cross product therefore acts on a circular pair as multiplication by iA_{z1} : color rotation becomes a $U(1)$ phase, and the 12 real color-2/3 fields close into 6 complex ones.

Physical picture. Color rotation is a phase. The background points along color-1, and rotations about that axis turn the (2, 3) color components into each other exactly like a planar rotation — i.e. like multiplying $X^2 + iX^3$ by a phase. The color-2/3 waves are therefore *charged* fields (charge ± 1) under the residual $U(1)$, and αA_{z1} acts on them like the z -component of an ordinary gauge potential: it shifts their momentum, $k_z \rightarrow k_z \pm \alpha A_{z1}$. All the non-Abelian structure of the problem is packed into this one “minimal-coupling” shift; the split into two different shifts (Ω_A in Ampère, Ω_F in Faraday) is the fingerprint of the model’s fixed $A_x = 0$ constraint, which treats the electric and magnetic covariant derivatives asymmetrically.

Define the two shifted wavenumbers that this produces,

$$\boxed{\Omega_A(\xi) = k_z + \alpha A_{z1} = k_z - u(\xi), \quad \Omega_F(\xi) = k_z - \alpha A_{z1} = k_z + u(\xi).} \quad (4.6)$$

4.2 Derivation of the six equations

Step 4.2.1 (Ampère- x). Write (3.5) out for the two colors $a = 2, 3$ separately:

$$\partial_t E_x^2 = -\partial_z B_y^2 - \alpha(\vec{A}_z \times \vec{B}_y)^2 - J_x^2, \quad \partial_t E_x^3 = -\partial_z B_y^3 - \alpha(\vec{A}_z \times \vec{B}_y)^3 - J_x^3. \quad (4.7)$$

Form (second equation) $\times i$ plus (first equation); by (4.1) the left side is $\partial_t(E_x^2 + iE_x^3) = \partial_t e_x$:

$$\partial_t e_x = -\partial_z(B_y^2 + iB_y^3) - \alpha[(\vec{A}_z \times \vec{B}_y)^2 + i(\vec{A}_z \times \vec{B}_y)^3] - (J_x^2 + iJ_x^3). \quad (4.8)$$

Take each of the three terms in turn.

- *Curl term.* $-\partial_z(B_y^2 + iB_y^3) = -\partial_z b$; the plane-wave ansatz $\partial_z \rightarrow ik_z$ turns this into $-ik_z b$.
- *Commutator term.* This is exactly the bracket evaluated in (4.5) with $\vec{X} = \vec{B}_y$ (so $X^c = b$): $(\vec{A}_z \times \vec{B}_y)^2 + i(\vec{A}_z \times \vec{B}_y)^3 = iA_{z1} b$. Multiplying by the overall $-\alpha$ gives $-i\alpha A_{z1} b$.
- *Current term.* From (3.9), $J_x^a = -n_A Q_A^a v_{xA} - n_B Q_B^a v_{xB}$. Its color-2,3 perturbation $\delta J_x^{2,3}$ is a sum of terms each containing either the background $v_{xA,B}$ (zero for both beams, (1.7) has no x -flow) or a product of two first-order perturbations ($\delta Q_s^{2,3} \cdot \delta v_{xs}$, itself second order and dropped in a linear theory). Hence $J_x^2 + iJ_x^3 = 0$ at linear order.

Collecting the two surviving terms and applying $\partial_t \rightarrow \gamma$:

$$\gamma e_x = -ik_z b - i\alpha A_{z1} b = -i(k_z + \alpha A_{z1}) b = -i\Omega_A b, \quad (4.9)$$

using $\Omega_A = k_z + \alpha A_{z1}$ from (4.6) in the last step.

Step 4.2.2 (Faraday). Write (3.7) for $a = 2, 3$ and combine as (2) + i (3) exactly as above, using $\partial_t(B_y^2 + iB_y^3) = \partial_t b$:

$$\partial_t b = \partial_x(E_z^2 + iE_z^3) - \partial_z(E_x^2 + iE_x^3) + \alpha[(\vec{A}_z \times \vec{E}_x)^2 + i(\vec{A}_z \times \vec{E}_x)^3]. \quad (4.10)$$

Term by term: $\partial_x(E_z^2 + iE_z^3) = \partial_x e_z \rightarrow e'_z$ (a prime denotes ∂_x , left as a derivative since the x -profile is not Fourier-expanded); $-\partial_z(E_x^2 + iE_x^3) = -\partial_z e_x \rightarrow -ik_z e_x$ under the ansatz; and the commutator term is (4.5) with $\vec{X} = \vec{E}_x$ (so $X^c = e_x$), giving $+i\alpha A_{z1} e_x$. Hence, with $\partial_t \rightarrow \gamma$,

$$\gamma b = -ik_z e_x + i\alpha A_{z1} e_x + e'_z = -i(k_z - \alpha A_{z1}) e_x + e'_z = -i\Omega_F e_x + e'_z, \quad (4.11)$$

using $\Omega_F = k_z - \alpha A_{z1}$ from (4.6). Note the opposite relative sign of the two αA_{z1} shifts in (4.9) vs (4.11): Ampère- x sees Ω_A , Faraday sees Ω_F . This sign difference has nothing to do with $\vec{X}$ being b versus e_x — (4.5) gives the *same* $+iA_{z1}X^c$ structure either way. It comes entirely from the overall sign in front of the commutator in the two model equations: (3.5) has “ $-\alpha(\dots)$ ” while (3.7) has “ $+\alpha(\dots)$ ”. That one model-level sign difference is the entire origin of $\Omega_A \neq \Omega_F$; their product, not either alone, is what enters the local dispersion relation (§5), and the split is inherited from the two distinct covariant structures in `khaxn.pdf` (cf. its eqs. (2)/(3) vs (16)/(19)).

Step 4.2.3 (Ampère- z). Write (3.6) for $a = 2, 3$ and combine as before:

$$\partial_t e_z = \partial_x(B_y^2 + iB_y^3) - (J_z^2 + iJ_z^3) = b' - (J_z^2 + iJ_z^3), \quad (4.12)$$

where $\partial_x B_y \rightarrow b'$ again denotes ∂_x (backward difference in the code; see §6.4); this equation has no commutator term at all, since the candidate A_x -term vanishes identically (§3.1). What remains is to linearize the current. From (3.9), for each beam s and color a , $J_z^a = -n_s Q_s^a v_{zs}$, summed over s ; each of n_s, Q_s^a, v_{zs} is background-plus-perturbation. Expand the product to first order for the color-2 piece of beam A (color-3 and beam B follow by relabelling):

$$\delta(n_A Q_A^2 v_{zA}) = \underbrace{(\delta n_A) \underbrace{Q_{A,0}^2}_{=0} v_{zA} + n_{A,0} (\delta Q_A^2) v_{zA} + n_{A,0} \underbrace{Q_{A,0}^2}_{=0} (\delta v_{zA})}_{=0}, \quad (4.13)$$

using that the background $Q_{A,B}^{2,3} = 0$ (only $Q^1 = \pm 1$ is nonzero in the background, (1.7)) kills the first and third terms; only the middle term, δQ_A^a multiplying the *background* velocity $v_{zA} = v$, survives. Repeating for beam B ($n_{B,0} = 1$, $v_{zB} = -v$, $Q_{B,0}^{2,3} = 0$) and assembling $\delta J_z^{2,3} = -\delta(n_A Q_A^{2,3} v_{zA}) - \delta(n_B Q_B^{2,3} v_{zB})$:

$$\delta J_z^{2,3} = -[n_A \delta Q_A^{2,3} v_{zA} + n_B \delta Q_B^{2,3} v_{zB}] = -v \delta Q_A^{2,3} - (-v) \delta Q_B^{2,3} = -v(\delta Q_A^{2,3} - \delta Q_B^{2,3}), \quad (4.14)$$

using $n_A = n_B = 1$, $v_{zA} = v$, $v_{zB} = -v$. Combining (2) + i(3) using $q_{A,B} \equiv \delta Q_{A,B}^2 + i\delta Q_{A,B}^3$ from (4.1) gives $J_z^2 + iJ_z^3 = -v(q_A - q_B)$. Substituting above, with $\partial_t \rightarrow \gamma$:

$$\gamma e_z = b' - [-v(q_A - q_B)] = b' + v(q_A - q_B). \quad (4.15)$$

Step 4.2.4 (Potential). Equation (3.8), $\partial_t A_z^a = -E_z^a$, is already linear and carries no commutator, so restricting to $a = 2, 3$ and combining (2) + i(3) is immediate: $\partial_t a = -e_z$, and with $\partial_t \rightarrow \gamma$,

$$\gamma a = -e_z. \quad (4.16)$$

Step 4.2.5 (Precession, beam A). Two pieces of setup, then the algebra.

Advective form and its linearization. As noted below (3.4), combined with continuity the density-weighted equation is equivalent to $dQ^a/dt = \alpha v_z (\vec{Q} \times \vec{A}_z)^a$ along a fluid trajectory, i.e. $\partial_t Q^a + v_z \partial_z Q^a = \alpha v_z (\vec{Q} \times \vec{A}_z)^a$ at this order ($v_x = 0$ in the background, so no x -advection of the background charge). Linearize about $Q_A^1 = 1$, $Q_A^{2,3} = 0$, $v_{zA} = v(\xi)$: the perturbation $\delta Q_A^{2,3}$ obeys $\partial_t \delta Q_A^{2,3} + v \partial_z \delta Q_A^{2,3} = \alpha v (\vec{Q} \times \vec{A}_z)^{2,3}$ (the source is already first order, so it needs no further linearization beyond what follows next). Under the plane-wave ansatz $\partial_z \rightarrow ik_z$ the advective term becomes $ik_z v \delta Q_A^{2,3}$: a Doppler shift by the beam's own local velocity.

The source, from the $SU(2)$ cross product. With $\vec{Q} = (1, Q^2, Q^3)$ (background color-1 plus color-2/3 perturbation) and $\vec{A}_z = (A_{z1}, a_2, a_3)$ (background plus perturbation), use $(\vec{X} \times \vec{Y})^2 = X^3 Y^1 - X^1 Y^3$, $(\vec{X} \times \vec{Y})^3 = X^1 Y^2 - X^2 Y^1$ (the general formula of (4.4), with now *both* vectors carrying nonzero components):

$$(\vec{Q} \times \vec{A}_z)^2 = Q^3 A_{z1} - Q^1 a_3 = Q^3 A_{z1} - a_3, \quad (4.17)$$

$$(\vec{Q} \times \vec{A}_z)^3 = Q^1 a_2 - Q^2 A_{z1} = a_2 - Q^2 A_{z1}, \quad (4.18)$$

using $Q^1 = 1$. Both expressions are already first order (each is a product of the background A_{z1} or the background $Q^1 = 1$ with a perturbation), as required to sit on the right of a linear equation. Combine as (2) + i(3), grouping the A_{z1} -terms and the a -terms separately:

$$(\vec{Q} \times \vec{A}_z)^2 + i(\vec{Q} \times \vec{A}_z)^3 = \underbrace{(Q^3 A_{z1} - iQ^2 A_{z1})}_{-iA_{z1}(Q^2 + iQ^3)} + \underbrace{(-a_3 + ia_2)}_{i(a_2 + ia_3)}. \quad (4.19)$$

Check each bracket by expanding it back out: $-iA_{z1}(Q^2 + iQ^3) = -iA_{z1}Q^2 + A_{z1}Q^3$, matching $Q^3 A_{z1} - iQ^2 A_{z1}$; and $i(a_2 + ia_3) = ia_2 - a_3$, matching $-a_3 + ia_2$. So, using $q_A \equiv Q^2 + iQ^3$ and $a \equiv a_2 + ia_3$ from (4.1),

$$(\vec{Q} \times \vec{A}_z)^c = ia - iA_{z1} q_A. \quad (4.20)$$

Assembling. The linearized precession equation reads $\gamma q_A + ik_z v q_A = \alpha v (\vec{Q} \times \vec{A}_z)^c = \alpha v (ia - iA_{z1} q_A)$. Move the Doppler term to the right and collect the two terms proportional to q_A :

$$\gamma q_A = i\alpha v a - ik_z v q_A - i\alpha v A_{z1} q_A = i\alpha v a - iv(k_z + \alpha A_{z1}) q_A = i\alpha v a - iv\Omega_A q_A, \quad (4.21)$$

using $\Omega_A = k_z + \alpha A_{z1}$ from (4.6) in the last step. The combination is meaningful: the beam's Doppler shift $k_z v$ and its precession frequency $\alpha v A_{z1}$ enter only through the single product $v\Omega_A$.

Step 4.2.6 (Precession, beam B). Repeat the previous step's two pieces with $Q_B^1 = -1$ and $v_{zB} = -v$.

Cross product. With $\vec{Q} = (-1, Q^2, Q^3)$, $\vec{A}_z = (A_{z1}, a_2, a_3)$:

$$(\vec{Q} \times \vec{A}_z)^2 = Q^3 A_{z1} - Q^1 a_3 = Q^3 A_{z1} + a_3, \quad (4.22)$$

$$(\vec{Q} \times \vec{A}_z)^3 = Q^1 a_2 - Q^2 A_{z1} = -a_2 - Q^2 A_{z1}, \quad (4.23)$$

where the sign flips relative to beam A come entirely from $Q_B^1 = -1$ replacing $Q_A^1 = 1$ in the terms $-Q^1 a_3$ and $Q^1 a_2$; nothing else changes. Combine as (2) + i(3):

$$(\vec{Q} \times \vec{A}_z)^2 + i(\vec{Q} \times \vec{A}_z)^3 = (Q^3 A_{z1} + a_3) + i(-a_2 - Q^2 A_{z1}) = \underbrace{(Q^3 A_{z1} - iQ^2 A_{z1})}_{-iA_{z1}(Q^2 + iQ^3)} + \underbrace{(a_3 - ia_2)}_{-i(a_2 + ia_3)}, \quad (4.24)$$

(check the second bracket: $-i(a_2 + ia_3) = -ia_2 + a_3$, matching $a_3 - ia_2$; the first bracket is identical to beam A's), so with $q_B \equiv Q^2 + iQ^3$,

$$(\vec{Q} \times \vec{A}_z)^c = -ia - iA_{z1} q_B. \quad (4.25)$$

Advective term. With background $v_{zB} = -v$, the Doppler term in the advective precession equation is $-v ik_z q_B$ (opposite sign from beam A because the background flow is reversed).

Assembling. $\gamma q_B - ik_z v q_B = \alpha(-v)(-ia - iA_{z1} q_B) = i\alpha v a + i\alpha v A_{z1} q_B$. Move the Doppler term to the right:

$$\gamma q_B = i\alpha v a + ik_z v q_B + i\alpha v A_{z1} q_B = i\alpha v a + iv(k_z + \alpha A_{z1}) q_B = i\alpha v a + iv \Omega_A q_B. \quad (4.26)$$

The two beams are driven identically ($+i\alpha v a$) but Doppler-rotate in opposite senses ($\mp iv \Omega_A$): both charge-sign flips (in the source and in the Doppler term) are compensated by the velocity sign flip except in the rotation term.

Collecting (4.9)–(4.26), the eigenvalue system $\gamma \Psi = L \Psi$, $\Psi = (b, e_x, e_z, a, q_A, q_B)$:

$$\boxed{\begin{array}{ll} \gamma b = -i\Omega_F e_x + e'_z & \gamma a = -e_z \\ \gamma e_x = -i\Omega_A b & \gamma q_A = i\alpha v a - iv \Omega_A q_A \\ \gamma e_z = b' + v(q_A - q_B) & \gamma q_B = i\alpha v a + iv \Omega_A q_B \end{array}} \quad (4.27)$$

4.3 Helicity conjugation and parity

The conjugate pairs $\bar{f} = f^2 - if^3$ satisfy (4.27) with $A_{z1} \rightarrow -A_{z1}$ (equivalently $u \rightarrow -u$, $\Omega_A \leftrightarrow \Omega_F$), because (4.5) acquires the opposite sign of iA_{z1} under conjugation. Separately, the background has the parity $v(-\xi) = -v(\xi)$, $A_{z1}(-\xi) = A_{z1}(\xi)$. Combining the two operations $\xi \rightarrow -\xi$ and helicity conjugation maps (4.27) onto itself with the same γ : *every eigenmode of one circular polarization localized on one side of the layer has a mirror partner of the opposite polarization on the other side, at exactly the same growth rate.* This is the symmetry behind the one-sided eigenfunctions found throughout the program (§6.3), and behind the helicity bookkeeping of the extraction pipeline (Appendix A).

4.4 Correspondence with the given frequency-domain theory; scope

Eliminating e_x , e_z , q_A , q_B , b from (4.27) (§6) produces beam response denominators $\gamma \pm iv \Omega_A$. Under $\omega = i\gamma$,

$$\gamma \pm iv \Omega_A = -i(\omega \mp v \Omega_A) = -i[(\omega \mp k_z v) \mp \alpha v A_{z1}], \quad (4.28)$$

which is exactly the structure $\Omega + \alpha \Phi_0^1$ of the given eq. (1.1) ($\Omega = \omega - k_z V_{0z}$ the Doppler-shifted frequency, $\Phi_0^1 = V_{0z} A_{0z}^1$ the effective background coupling), one factor per beam. Likewise the left-hand operator of (1.1) contains $k_z^2 - \omega^2 - \alpha^2 A_{0z}^2 = \gamma^2 + k_z^2 - u^2 / \dots$, the factor $\mathcal{D}$ of §6. The time-domain model therefore linearizes into precisely the given theory, with two deliberate scope differences:

1. no background scalar potential / background electric field ($\phi_0^1 = E_{0x}^1 = 0$ in the simulation; the $\alpha^2(\phi_0^1)^2$ term of (1.1) is absent) — a genuine physical truncation, *unrelated* to the gauge-counting question resolved in §3.2: nothing in the gauge freedom requires ϕ_0 to vanish, and this is the one place the given theory’s background is actually cut down, rather than merely repackaged; and
2. no background B_{y1} coupling in Ampère- x (the same term identified and dropped on geometric grounds in the energy budget, eq. (3.30) of §3.5) and no fluid density/momentum perturbations in the state vector (their combined effect on the shear branch is bounded at the few-percent level by the simulation-vs-solver agreement); and the $a_y^\pm$ sector of the given theory (its eq. 22, decoupled from a_z already at the level of `khaxn.pdf`’s own final equations) is not simulated at all — a sector choice, not a further gauge-fixing, as established in §3.2.

5 Local dispersion: the tachyonic outer branch and the local drive branch

Here we freeze the background locally ($A_{z1} = A$, v constant) and take plane waves in x ($\partial_x \rightarrow ik_x$). The system (4.27) becomes a 6×6 matrix problem; at $k_x = 0$ it factorizes into two independent blocks whose exact dispersion relations we now derive.

5.1 Block factorization at $k_x = 0$

Setting $e'_z \rightarrow ik_x e_z = 0$ and $b' \rightarrow ik_x b = 0$ in (4.27): the pair (b, e_x) closes on itself,

$$\gamma b = -i\Omega_F e_x, \quad \gamma e_x = -i\Omega_A b \quad (\text{EM block}), \quad (5.1)$$

and the quadruple (e_z, a, q_A, q_B) closes on itself,

$$\gamma e_z = v(q_A - q_B), \quad \gamma a = -e_z, \quad \gamma q_{A,B} = i\alpha v a \mp i v \Omega_A q_{A,B} \quad (\text{charge block}). \quad (5.2)$$

5.2 EM block: the tachyonic branch

Step 5.2.1. Multiply the two equations of (5.1):

$$\gamma^2 b e_x = (-i\Omega_F e_x)(-i\Omega_A b) = -\Omega_A \Omega_F b e_x \implies \gamma^2 = -\Omega_A \Omega_F = \alpha^2 A^2 - k_z^2. \quad (5.3)$$

Step 5.2.2 (Interpretation). The two circular polarizations carry adjoint color charge ± 1 under the background color-1 rotation; the covariant z -derivative $D_z = \partial_z \mp i\alpha A_{z1}$ gives them effective wavenumbers $k_z \pm \alpha A$. The product $\Omega_A \Omega_F = (k_z - u)(k_z + u) = k_z^2 - u^2$ is negative wherever the background shift exceeds the wavenumber, and there one polarization becomes *tachyonic*: $\gamma^2 > 0$. This is the charged-vector-on-strong-color-background instability of the Nielsen–Olesen class (Nielsen & Olesen 1978), realized here in 2D with A playing the role of the strong background field.

Step 5.2.3 (Instability region and local envelope). With the production background (1.8), $u(\xi) = \alpha V_0 \ln \cosh \xi$ and (1.9), the local growth rate

$$\gamma_{\text{loc}}(\xi) = \sqrt{u(\xi)^2 - k_z^2} \simeq \sqrt{(\alpha V_0)^2 (|\xi| - \ln 2)^2 - k_z^2} \quad (5.4)$$

is real exactly for

$$\boxed{|\xi| > \xi_{\text{crit}} \simeq \frac{k_z}{\alpha V_0} + \ln 2} \quad (5.5)$$

and grows asymptotically linearly, $\gamma_{\text{loc}} \rightarrow \alpha V_0 |\xi|$ (Fig. 3).

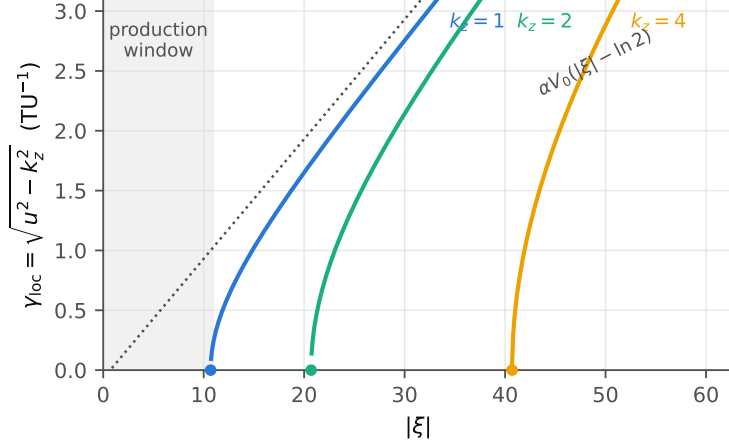


Figure 3: The tachyonic local envelope (5.4) at $\alpha = 2$, $V_0 = 0.05$ for three k_z . Dots on the axis mark the critical radii (5.5); the dotted line is the linear asymptote $\alpha V_0(|\xi| - \ln 2)$. Higher k_z pushes the instability outward — which is why a production window (shaded, $\xi_w = 11$) can exclude the branch entirely for the measured modes, and why the exclusion fails at large α (Corollary 5.2).

Step 5.2.4 (Finite k_x only subtracts). Keep $k_x \neq 0$ but drop the charge coupling (outer region, v at its plateau; the charge terms decouple from the growing branch — verified numerically at all outer eigenmode peaks). Then (4.27) retains e_z : from $\gamma e_z = ik_x b$, $\gamma b = -i\Omega_F e_x + ik_x e_z$, $\gamma e_x = -i\Omega_A b$:

$$\gamma^2 b = -i\Omega_F(\gamma e_x) + ik_x(\gamma e_z) = -\Omega_A \Omega_F b - k_x^2 b \implies \gamma^2 = u^2 - k_z^2 - k_x^2. \quad (5.6)$$

$\gamma_{\text{loc}}(\xi)$ of (5.4) is therefore the *pointwise envelope* over all x -structures; global modes must oscillate ($k_x^2 > 0$ costs growth) and sit below it.

Physical picture. An effective wavenumber crossing zero. A wave with $k_{z,\text{eff}}^2 = k_z^2 - u^2 < 0$ has an imaginary wavenumber — as in an overdense plasma, but with the roles inverted: it is the *restoring* term of the wave operator that changes sign, so instead of evanescing in space the mode grows in time, rolling down an inverted potential. And it is the same shift u that, inside ξ_{crit} , deepens the confining well of the shear branch (through $\mathcal{D}$ in §6): one term, two fates, split at ξ_{crit} . Window design is nothing more than choosing which side of ξ_{crit} the simulation keeps.

Remark 5.1 (Energy source and the freeze artifact). By (3.35), this branch is powered entirely by the frozen A_{z1} ; the model as posed makes it a true linear instability with a bottomless supply. Its runaway character (not its existence) is the freeze artifact; windows (sponge/cut) placed inside ξ_{crit} are the legitimate excision of this branch from shear-physics measurements. The self-consistent test (§13.2) shows the unfrozen back-reaction does not deplete the background either — it deepens it locally and stabilizes only weakly and globally.

Corollary 5.2 (High- α window collapse). *A window of radius ξ_w excludes the tachyonic branch only if $\xi_w < \xi_{\text{crit}} = k_z/(\alpha V_0) + \ln 2$. As α grows at fixed k_z, V_0 , $\xi_{\text{crit}} \rightarrow \ln 2$: eventually no window can both contain the shear-branch well (which needs ξ_w of order several) and exclude the tachyon. This is the mechanism of the early tachyonic blow-ups in the $\alpha \geq 6$ corner of the EPS-scan $v2$ campaign: at $\alpha = 6$, $V_0 = 0.05$, $k_z = 2$, $\xi_{\text{crit}} = 7.4$, already inside the vetted windows, and the failure maps to α , not k_z , exactly as (5.5) predicts.*

Remark 5.3 (The “precession cascade” scale $\gamma_{\text{casc}} \approx \alpha V_0$). Near the layer ($|\xi| = \mathcal{O}(1)$, $|A_{z1}| \lesssim V_0$), the same color-mixing terms rotate the color-2/3 sector at the local rate $\alpha|A_{z1}| \sim \alpha V_0$. Seeding campaigns observed the resulting build-up of $A_z^{2,3}$ at $\gamma_{\text{casc}} \approx \alpha V_0$ (0.20–0.24 at $\alpha = 2$, $V_0 = 0.1$),

which can mask slower KH growth — the motivation for eigenfunction seeding (§15). It is the same mechanism as (5.3) evaluated at the layer-scale $|A_{z1}|$ rather than the outer plateau.

5.3 Charge block: the local drive branch

Step 5.3.1 (Solve the two precession equations). From (5.2),

$$q_A = \frac{i\alpha v a}{\gamma + iv\Omega_A}, \quad q_B = \frac{i\alpha v a}{\gamma - iv\Omega_A}. \quad (5.7)$$

Step 5.3.2 (Beam-difference response).

$$q_A - q_B = i\alpha v a \left[\frac{1}{\gamma + iv\Omega_A} - \frac{1}{\gamma - iv\Omega_A} \right] = i\alpha v a \frac{-2iv\Omega_A}{\gamma^2 + v^2\Omega_A^2} = \frac{2\alpha v^2\Omega_A}{\mathcal{R}} a, \quad \mathcal{R} \equiv \gamma^2 + v^2\Omega_A^2. \quad (5.8)$$

Step 5.3.3 (Close through e_z and a). $\gamma e_z = v(q_A - q_B) = \frac{2\alpha v^3\Omega_A}{\mathcal{R}} a \equiv g a$ and $e_z = -\gamma a$ give

$$-\gamma^2 a = g a \quad \Rightarrow \quad \boxed{\gamma^2 = -g, \quad g(\xi; \gamma) \equiv \frac{2\alpha v^3\Omega_A}{\gamma^2 + v^2\Omega_A^2}.} \quad (5.9)$$

On the side of the layer where $v^3\Omega_A < 0$, $g < 0$ and the block grows: this is the *local (homogeneous) limit of the shear-branch drive*. The function g defined here is exactly the drive that appears in the reduced ODE (§6); equation (5.9) is its $k_x \rightarrow 0$ limit, and its maximization over the background will produce the growth ceiling in §7.

Physical picture. The charge block is a pair of driven pendulums. The potential a shakes each beam's color charge (the $i\alpha v a$ source); each charge then precesses in the background at its own Doppler-shifted frequency $\pm v\Omega_A$ (opposite for the two beams). Their *difference* is what radiates, through the current $-v(q_A - q_B)$, and it feeds a back with a sign set by which beam leads in phase. Growth is a phase-locking phenomenon between the wave and the beams' color precession — the resonance structure this implies is made exact in §7.

5.4 Airy estimate for wall-bounded outer modes

Global tachyonic modes live in the annulus between ξ_{crit} and the window radius ξ_w (hard cut: Dirichlet wall). Their rates follow from a one-dimensional bound-state problem in the local potential $U(x) \equiv \gamma_{\text{loc}}^2(x)$, increasing outward:

$$a'' + (U(x) - \gamma^2)a = 0, \quad (5.10)$$

with $a = 0$ at the wall x_w and decay toward the layer.

Step 5.4.1 (Linearize at the wall). Near x_w write $U(x) = U_w - U'(x_w - x)$ with $U_w = U(x_w)$, $U' = dU/dx > 0$. In the inward coordinate $t = x_w - x \geq 0$:

$$\frac{d^2 a}{dt^2} - (\gamma^2 - U_w + U' t)a = 0. \quad (5.11)$$

Step 5.4.2 (Airy form). Substitute $\tau = (U')^{1/3}t + \tau_0$ with $\tau_0 = (\gamma^2 - U_w)/(U')^{2/3}$:

$$\frac{d^2 a}{d\tau^2} = \tau a, \quad (5.12)$$

whose solution decaying for $\tau \rightarrow +\infty$ (deep inside, where $U < \gamma^2$) is $\text{Ai}(\tau)$.

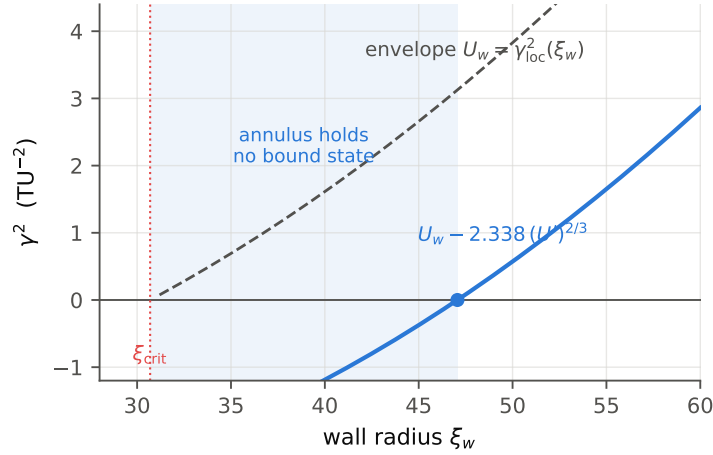


Figure 4: The Airy wall estimate (5.13) for outer modes bounded by a hard cut at ξ_w ($\alpha = 1$, $V_0 = 0.05$, $k_z = 1.5$, $\varepsilon_s = 0.15$). The dashed curve is the pointwise envelope $U_w = \gamma_{\text{loc}}^2(\xi_w)$; the solid curve subtracts the quantization cost $2.338 (U')^2/3$ of bending the mode to zero at the wall. In the shaded range of wall radii the corrected γ^2 is negative: the annulus between ξ_{crit} and the wall is too narrow to hold any bound state, which is why tight windows remove the outer branch outright rather than merely weakening it.

Step 5.4.3 (Wall condition). The Dirichlet wall at $t = 0$ requires $\text{Ai}(\tau_0) = 0$. The least negative zero of Ai is $-a_1 = -2.338\dots$, so the deepest (fastest) mode has $\tau_0 = -a_1$:

$$\boxed{\gamma^2 \simeq U_w - 2.338 (U')^2/3} \quad (5.13)$$

Corollary 5.4 (Why tight windows kill the branch outright). *For a tight window, $U_w = \gamma_{\text{loc}}^2(\xi_w)$ is small (the wall sits just past ξ_{crit}) while the slope $U' \simeq 2(\alpha V_0)^2(\xi_w - \ln 2)/\varepsilon_s$ is not; (5.13) then goes negative: the annulus holds no bound state at all, and windowing does not merely reduce the outer branch — it removes it (Fig. 4). Measured global rates at the widest tested annuli reach $0.78\times$ the envelope with a discrete ladder below, in line with the quantization cost in (5.13).*

6 Exact scalar reduction of the six-field system

The six-field system (4.27) contains only one x -derivative chain ($e'_z \rightarrow b \rightarrow b'$); everything else is algebraic. It therefore reduces *exactly* to one second-order ODE for the potential amplitude $a(x)$. This reduction, verified to machine precision against the ARPACK eigenvalues of the discrete six-field operator, is the backbone of the exact-action theory.

6.1 Continuum elimination

Step 6.1.1 (Eliminate e_x). From (4.9): $e_x = -i\Omega_A b/\gamma$.

Step 6.1.2 (Eliminate e_z). From (4.16): $e_z = -\gamma a$.

Step 6.1.3 (Express b through a'). Insert both into Faraday (4.11):

$$\gamma b = -i\Omega_F \left(-\frac{i\Omega_A}{\gamma} b \right) + (-\gamma a)' = -\frac{\Omega_A \Omega_F}{\gamma} b - \gamma a'. \quad (6.1)$$

Multiply by γ and collect b :

$$b(\gamma^2 + \Omega_A \Omega_F) = -\gamma^2 a' \quad \implies \quad b = -\frac{\gamma^2 a'}{\mathcal{D}}, \quad \boxed{\mathcal{D}(x) \equiv \gamma^2 + \Omega_A \Omega_F = \gamma^2 + k_z^2 - u(x)^2} \quad (6.2)$$

$\mathcal{D}$ is the electromagnetic factor: $\mathcal{D} < 0$ is exactly the tachyonic region of §5.2 (one polarization sees a negative effective k^2); the shear branch lives where $\mathcal{D} > 0$.

Step 6.1.4 (Charge response). Equations (5.7)–(5.8) hold pointwise with $v = v(\xi)$, $\Omega_A = \Omega_A(\xi)$:

$$q_A - q_B = \frac{2\alpha v^2 \Omega_A}{\mathcal{R}} a, \quad \mathcal{R}(x) = \gamma^2 + v^2 \Omega_A^2. \quad (6.3)$$

Step 6.1.5 (Assemble Ampère- z). Insert $e_z = -\gamma a$, (6.2) and the charge response into (4.15):

$$-\gamma^2 a = \left(-\frac{\gamma^2 a'}{\mathcal{D}} \right)' + \frac{2\alpha v^3 \Omega_A}{\mathcal{R}} a \quad (6.4)$$

and rearrange:

$$\boxed{\left(\frac{\gamma^2 a'}{\mathcal{D}} \right)' = (\gamma^2 + g) a, \quad g(x) = \frac{2\alpha v^3 \Omega_A}{\gamma^2 + v^2 \Omega_A^2}} \quad (6.5)$$

— one Sturm–Liouville-type equation, nonlinear in the eigenvalue γ (which appears in $\mathcal{D}$, g , and the right side), but scalar. (“Sturm–Liouville” is only the mathematicians’ name for the flux-form structure $(p a')' = W a$ that the stationary Schrödinger equation also has; nothing deeper is implied by the term. The next subsection unpacks the analogy at length.)

6.2 How to read the reduced equation: a dictionary for the non-specialist

Equation (6.5) is the pivot of the entire document: every result that follows — the “drive”, the “resonance”, the “Lorentzian” beam response, the growth “ceiling” (§7), the “well” and its “quantization” (§8) — is a statement about this one ODE. That vocabulary is second nature in plasma physics and quantum mechanics but is jargon to everyone else, and until now it has not been earned by the text. This subsection therefore builds each term explicitly from three pieces of a standard bachelor’s curriculum: the finite potential well, the driven damped oscillator, and spin precession in a magnetic field. Readers already fluent in the vocabulary can skip to §6.3 without loss — nothing here is used as a technical ingredient later, only as language.

The reduced equation is a Schrödinger eigenvalue problem in disguise. Recall the stationary Schrödinger equation for a particle in a one-dimensional potential, written with everything on one side:

$$\psi''(x) = \frac{2m}{\hbar^2} [V(x) - E] \psi(x). \quad (6.6)$$

Its qualitative behavior is controlled entirely by the sign of the bracket. Where $E > V$ (the *classically allowed* region) ψ oscillates in x ; where $E < V$ (*forbidden* region) ψ grows or decays exponentially. A *well* is an allowed region sandwiched between forbidden ones; a *bound state* is a solution that decays into the forbidden regions on both sides; and requiring the oscillation inside the well to join smoothly onto decay outside is possible only at discrete values of E — that matching condition is all that *quantization* means, here and in every later section.

Equation (6.5) has exactly this structure, with two differences. First, the second derivative appears in “flux form”, $(\gamma^2 a' / \mathcal{D})'$, with a position-dependent coefficient — the same form the Schrödinger equation takes for a position-dependent mass, or the steady heat-conduction equation for a position-dependent conductivity. This changes bookkeeping (the $\mathcal{D}$ factors tracked in §6.3) but not the oscillate-versus-decay dichotomy. The precise analogue of the bracket $\frac{2m}{\hbar^2} [V - E]$ — the single function of x whose sign separates oscillation from decay — is written down in §6.3, eq. (6.10), as $-Q(x; \gamma)$. Second, and less trivially, the eigenvalue γ does not enter linearly the way E does: it is buried inside $\mathcal{D}$, inside g , and on the right-hand side. In practice such a problem is solved iteratively (guess γ , solve the resulting *linear* well problem, correct γ , repeat — this is what the Newton iteration of §6.4 does), and none of the well/bound-state/quantization reasoning is affected.

What is being solved for. Every perturbation was assumed to evolve as $e^{\gamma t}$ (§4), so a solution with $\gamma > 0$ is an exponentially growing wave — an instability. The role of the energy eigenvalue E is played by the growth rate: instead of asking “which energies admit a bound state?” one asks “which growth rates admit a trapped, self-consistently growing wave profile?”. The largest such γ is what the simulation measures, because after a few e -foldings the fastest exponential dominates regardless of how the initial condition distributed amplitude among the modes.

What “drive” means. Look at the right-hand side of (6.5), $(\gamma^2 + g)a$. If g were zero, the bracket would be positive everywhere, and (wherever $\mathcal{D} > 0$) the equation would admit only exponential x -profiles — forbidden region everywhere, no trapped mode, no instability. A well can open only where g is *negative and exceeds γ^2 in magnitude*. So g is the one term that makes growth possible: it measures how strongly the beams’ collective response feeds energy back into the wave, and it is called the *drive* in exactly the sense of the driving force of a forced oscillator. Its microscopic origin is the color current $\delta J_z = -v(q_A - q_B)$ of the beams’ perturbed charges, which entered Ampère- z in (4.15); a growing mode is a wave that digs its own potential well out of the beam response and then lives in it.

Where the response denominator $\mathcal{R}$ comes from: a driven-precession toy problem. The building block of g is the beam response (5.7), and it is worth deriving once in stripped-down notation. Let a single complex variable $q(t)$ obey

$$\frac{dq}{dt} = -i\Omega_{\text{prec}} q + S e^{\gamma t}. \quad (6.7)$$

With $S = 0$ the solution is $q \propto e^{-i\Omega_{\text{prec}} t}$: free rotation of the complex phase at frequency Ω_{prec} . This is mathematically identical to a magnetic moment precessing about a static magnetic field (Larmor precession); in our problem the background potential plays the role of the field and the beam’s color charge plays the role of the moment — the precession equations (4.21)–(4.26) are precisely of this form with $\Omega_{\text{prec}} = \pm v\Omega_A$, the sign differing between the beams because they stream in opposite directions. Now switch on the forcing $S e^{\gamma t}$ (the wave a shaking the charge) and look for the driven response $q = C e^{\gamma t}$:

$$\gamma C = -i\Omega_{\text{prec}} C + S \implies C = \frac{S}{\gamma + i\Omega_{\text{prec}}}, \quad |C|^2 = \frac{|S|^2}{\gamma^2 + \Omega_{\text{prec}}^2}. \quad (6.8)$$

This two-line calculation *is* (5.7), and the beam-difference algebra of (5.8) then produces the denominator $\mathcal{R} = \gamma^2 + v^2\Omega_A^2$ that recurs everywhere afterwards. Note the role played by γ : it sits where a damping rate would sit in the classical driven-oscillator formula. Not because anything is dissipated — the model is dissipation-free — but because a signal growing as $e^{\gamma t}$ is not a pure sinusoid: over one e -folding time $1/\gamma$ it cannot distinguish precession phases within a band of width $\sim \gamma$, so the response is smeared over that bandwidth exactly as damping smears a classical resonance.

“Lorentzian” is the name of a curve, not of a group. The bell-shaped function

$$L(\Omega) = \frac{1}{\gamma^2 + \Omega^2} \quad (6.9)$$

— peak height $1/\gamma^2$ at $\Omega = 0$, half-width γ — is called a *Lorentzian*, after H. A. Lorentz’s classical model of spectral-line broadening. It is the universal response curve of a driven system with finite bandwidth: the resonance curve of the damped driven oscillator, the shape of atomic and laser spectral lines, the Breit–Wigner resonance shape of particle physics, the NMR absorption line. Its odd companion $\Omega L(\Omega) = \Omega/(\gamma^2 + \Omega^2)$, with extrema at $\Omega = \pm\gamma$, is the standard *dispersive* line shape of resonance physics. When §7 says “the beam response is a Lorentzian in the local

precession frequency”, it means precisely that $1/\mathcal{R}$ is the curve L evaluated at $\Omega = v\Omega_A$; the drive g , which carries one extra factor of the precession frequency (the beam *difference* of (5.8) vanishes at $\Omega_A = 0$), is the dispersive companion. The word has nothing to do with Lorentz transformations, boosts, or the Lorentz group.

Physical picture. The whole instability in one paragraph. The wave a shakes each beam’s color charge (the source $iav a$ in (4.21)); each charge precesses about the color-1 background like a magnetic moment about a static field, at the Doppler-shifted rate $\pm v\Omega_A$; because the beams stream oppositely they precess in opposite senses, so their responses differ; the difference is a net color current $\delta J_z = -v(q_A - q_B)$; and that current feeds back into the wave through Ampère’s law. Where the feedback arrives with the right phase the loop has gain ($g < 0$), the wave digs its own well and grows inside it. Everything that follows quantifies this loop: §6.3 reads off which factors control the gain, §7 finds where the gain is largest (a resonance) and how large it can possibly be (the ceiling $\gamma^3 \leq \alpha V_0^2$), and §8 determines which wave shapes fit inside the well (quantization).

6.3 Anatomy of the reduced equation

With the dictionary of §6.2 in hand, eq. (6.5) can be read factor by factor.

Physical picture. Why three powers of v . The wave must, in order: (i) *shake* the charges — only moving beams feel the precession source, one factor of v ; (ii) make the two beams respond *differently* — otherwise their currents cancel, and the only asymmetry between them is their opposite Doppler rotation, a second factor of v ; (iii) *collect* the response as a current, $\delta J_z = -v(q_A - q_B)$, a third factor of v . The drive therefore dies as v^3 toward the layer center: the shear branch is anchored to the *flow*, not to the field maximum. Its one-sidedness is helicity selection — each circular polarization couples to the side whose flow direction matches its rotation sense, and the mirror mode belongs to the conjugate polarization (§4.3).

- **$\mathcal{D}$ (EM factor).** Sign changes of $\mathcal{D}$ separate the propagating/trapped shear-branch region from the tachyonic outer region: one mechanism — the covariant shift u — creates both the confining well and, past $|\xi| = \xi_{\text{crit}}$, the runaway. (“The well and the tachyon are two sides of one term.”)
- **g (drive).** The *only* drive of the shear branch, and precisely the local-drive function of (5.9). Its three powers of v have distinct origins: one from the precession source ($iava$ in (4.21)), one from the beam Doppler asymmetry (the $v\Omega_A/\mathcal{R}$ structure of (5.8)), one from the current ($\delta J_z = -v(q_A - q_B)$). Counter-streaming enters only through the even combination $\mathcal{R} = \gamma^2 + v^2\Omega_A^2$.
- **One-sidedness.** $g \propto v^3$ is odd in ξ (for $\Omega_A > 0$): each circular polarization is driven on one side of the layer only; the mirror image is carried by the conjugate polarization at the same $|\gamma|$ (§4.3). This explains the one-sided eigenfunctions seen in all solver outputs.
- **WKB form.** Writing $a \propto \exp(\pm i \int \sqrt{Q} dx)$ brings (6.5) to the standard form with the exact local wavenumber

$$Q(x; \gamma) = - \frac{\mathcal{D}(\gamma^2 + g)}{\gamma^2} \quad (6.10)$$

(the flux form contributes the $\mathcal{D}$ weight; a well requires $Q > 0$, i.e. $\mathcal{D} > 0$ together with $g < -\gamma^2$).

Figure 5 displays $\mathcal{D}$ and Q at a production point, computed at the model-B eigenvalue.

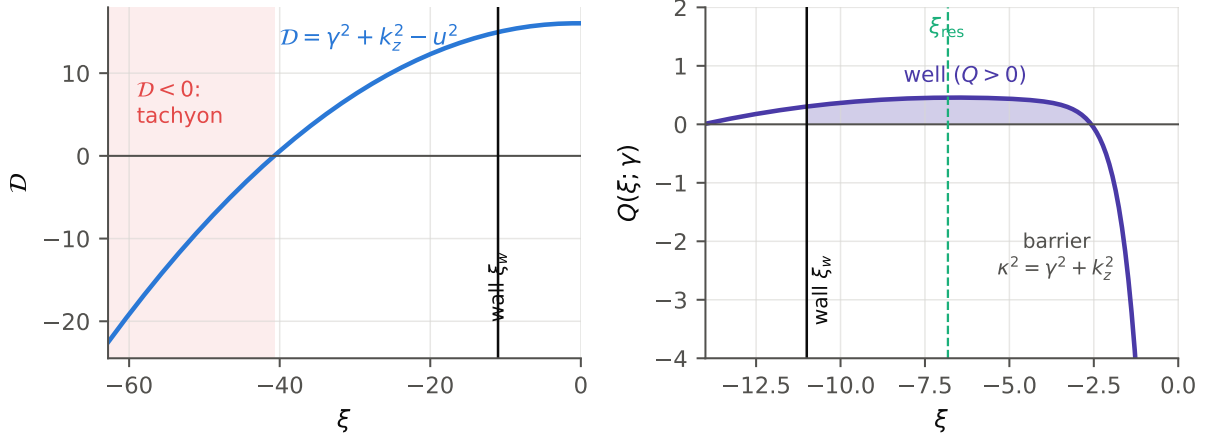


Figure 5: Anatomy of the reduced equation (6.5) on the driven side ($\alpha = 2$, $V_0 = 0.05$, $\varepsilon_s = 0.15$, $k_z = 4$, cut $\xi_w = 11$, $\gamma = 0.1694$ from model B). *Left*: the EM factor $\mathcal{D}$ over the whole half-box; $\mathcal{D} < 0$ (shaded) beyond $\xi_{\text{crit}} \approx -41$ is the tachyonic region — far outside the wall, which is the point of the window. *Right*: the exact local wavenumber $Q(\xi; \gamma)$ of (6.10) near the layer: the shallow well ($Q > 0$, shaded) between the wall and the $\tanh^3$ knee, the resonance surface ξ_{res} of (7.10) inside it, and the strong evanescent barrier ($Q \rightarrow -(\gamma^2 + k_z^2)$, off scale) toward the layer. The shallowness of the well — it holds less than a quarter wavelength — is the quantization marginality of §8.

6.4 Exactness at the discrete level; the two wall faces

The eigensolver uses a forward difference for $\partial_x e_z$ (Faraday row) and a backward difference for $\partial_x b$ (Ampère- z row), matching the simulation’s Yee stagger. Repeat the elimination on the lattice ($x_i = i \, dx$):

Step 6.4.1. The discrete Faraday/Ampère- x pair gives, per node i ,

$$b_i \frac{\mathcal{D}_i}{\gamma} = \frac{e_{z,i+1} - e_{z,i}}{dx} \implies b_i = -\frac{\gamma^2}{\mathcal{D}_i} \frac{a_{i+1} - a_i}{dx}, \quad (6.11)$$

i.e. b_i lives on the *face* between nodes i and $i+1$, with $\mathcal{D}$ evaluated at the left node of the face.

Step 6.4.2. The discrete Ampère- z row then reads

$$-\gamma^2 a_i = \frac{b_i - b_{i-1}}{dx} + g_i a_i = -\frac{\gamma^2}{dx^2} \left[\frac{a_{i+1} - a_i}{\mathcal{D}_i} - \frac{a_i - a_{i-1}}{\mathcal{D}_{i-1}} \right] + g_i a_i, \quad (6.12)$$

the exact flux-form finite difference of (6.5). No approximation was made: the $6N \times 6N$ eigenproblem and the $N \times N$ nonlinear problem (6.12) have identical spectra on the shared modes. Verified: Newton iteration on (6.12) reproduces the six-field ARPACK eigenvalues to relative difference 0.0 (below 10^{-10}) at seven points spanning $\alpha = 1\text{--}3$, $V_0 = 0.03\text{--}0.1$, $k_z = 1\text{--}7$.

Step 6.4.3 (Wall faces). A hard cut (`xi_cut`) zeroes all fields outside radius ξ_w . In the reduction the two sides of the wall behave differently:

- where the *right* neighbor is outside, $e_{z,i+1}$ is a Dirichlet ghost ($e_z = 0$, hence $a = 0$): the face flux uses a zero value — a Dirichlet face;
- where the *left* neighbor is outside, b_{i-1} is dropped: the face flux itself vanishes — a zero-flux (Neumann, $b = 0 \equiv a' = 0$) face.

The mode side of the wall is therefore Neumann for a , the far side Dirichlet; the two faces are *not* equivalent, and the Neumann condition on the mode side is what enters the quantization phase in §8.

7 The drive: precession resonance, the growth ceiling, and natural units

The plan of this section, in plain language. §6 ended with a wave trapped in a well whose depth is set by the drive g , together with the criterion (below (6.10)) that a well exists only where $-g > \gamma^2$. Two elementary questions therefore control everything. *Where*, as a function of position, is the drive strongest? Answered in §7.2 by maximizing a function of one variable: on a *resonance surface*, where the beam-frame precession frequency matches the growth rate. *How strong* can the drive possibly be? The maximum turns out to be $\alpha V_0^2/\gamma$, and inserting it into the well-existence criterion bounds the growth rate of *every* mode by $\gamma^3 \leq \alpha V_0^2$ — the ceiling, which the measured dispersion peaks saturate to within a few percent. No new physics enters this section: it is first-year calculus applied to the drive g of (6.5).

7.1 The saturated drive $G(w)$

Step 7.1.1 (Specialize g to the saturation plateaus). The drive $g(x) = 2\alpha v^3 \Omega_A / (\gamma^2 + v^2 \Omega_A^2)$ of (6.5) contains two background profiles, $v(\xi)$ and $\Omega_A(\xi) = k_z - u(\xi)$. For $|\xi| \gtrsim 2$ both simplify: the shear is saturated, $v = \pm V_0$ (so $v^2 = V_0^2$ exactly), and u grows linearly, $u = \alpha V_0(|\xi| - \ln 2)$ by (1.9). All position dependence of g then enters through the single combination

$$w \equiv \Omega_A(\xi) = k_z - u(\xi), \quad (7.1)$$

the *local covariant wavenumber*: the wavenumber the wave effectively carries at position ξ after the background's minimal-coupling shift (§4). At the layer $u \approx 0$, so $w \approx k_z$; moving outward, w decreases linearly with $|\xi|$ at slope αV_0 , crossing zero at $|\xi| = \xi_{\text{crit}}$ (5.5). On each plateau, position and w are thus interchangeable coordinates, and any statement about the best w translates directly into the position ξ where that w is attained (§7.4).

Step 7.1.2 (Sign bookkeeping: which side of the layer drives). A well needs $g < 0$ (§6.2), and the sign of g is the sign of $v^3 \Omega_A$. Between the layer and ξ_{crit} , where $w = \Omega_A > 0$, this requires $v < 0$: for the $2 + i3$ polarization the driven side is $\xi < 0$. (The other side is not lost — by the helicity–parity symmetry of §4.3 it carries the mirror mode of the conjugate polarization at exactly the same growth rate, so studying one side loses no generality.) On the driven side $v = -V_0$, hence $v^3 = -V_0^3$, and the *magnitude* of the drive is

$$-g = G(w) \equiv \frac{2\alpha V_0^3 w}{\gamma^2 + V_0^2 w^2}, \quad w > 0. \quad (7.2)$$

Step 7.1.3 (Shape of G : weak when slow, weak when fast). G is the dispersive Lorentzian of §6.2 in the precession frequency $V_0 w$, and its two limits have transparent physical readings:

- $V_0 w \ll \gamma$ (slow precession): $G \simeq (2\alpha V_0^3/\gamma^2) w \rightarrow 0$. At $w = 0$ the two beams' precession frequencies $\pm V_0 w$ both vanish, the beams respond identically, and their currents cancel ($q_A - q_B \rightarrow 0$ in (5.8)): no beam asymmetry, nothing to feed the wave.
- $V_0 w \gg \gamma$ (fast precession): $G \simeq 2\alpha V_0/w \rightarrow 0$. The charges now turn through many radians during one e -folding time $1/\gamma$ of the wave; their response averages out over the fast rotation and little net work is done.

Between the two limits G must peak — the resonance, located next.

7.2 Resonance and the exact ceiling

Step 7.2.1 (Maximize G over w).

$$\frac{dG}{dw} = 2\alpha V_0^3 \frac{(\gamma^2 + V_0^2 w^2) - w(2V_0^2 w)}{(\gamma^2 + V_0^2 w^2)^2} = 2\alpha V_0^3 \frac{\gamma^2 - V_0^2 w^2}{(\gamma^2 + V_0^2 w^2)^2} = 0 \implies w^* = \frac{\gamma}{V_0}. \quad (7.3)$$

The condition $V_0 w^* = \gamma$ says: the mode grows fastest where the *beam-frame precession frequency* $v\Omega_A = V_0 w$ matches the growth rate — a precession resonance surface in the flow. The peak value is

$$G_{\max} = G(w^*) = \frac{2\alpha V_0^3 (\gamma/V_0)}{2\gamma^2} = \frac{\alpha V_0^2}{\gamma}. \quad (7.4)$$

Step 7.2.2 (Ceiling, derivation 1 (well existence)). By (6.10), a confining well ($Q > 0$ with $\mathcal{D} > 0$) requires $g < -\gamma^2$, i.e. $G > \gamma^2$ somewhere. Since $G \leq G_{\max}$,

$$\gamma^2 \leq \frac{\alpha V_0^2}{\gamma} \iff \boxed{\gamma^3 \leq \alpha V_0^2} \quad (7.5)$$

independent of k_z , ε_s , and window geometry (Fig. 6, left).

Physical picture. A self-limiting resonance. The beam response is a Lorentzian (the resonance line shape of §6.2) in the local precession frequency whose width is set by the growth rate itself ($\mathcal{R} = \gamma^2 + v^2\Omega_A^2$): the faster the mode grows, the broader the band of precession frequencies it averages over — exactly as a larger damping broadens and flattens a classical resonance curve — and the weaker the peak drive $G_{\max} = \alpha V_0^2/\gamma$ it can extract. The ceiling $\gamma^3 = \alpha V_0^2$ is the fixed point of this self-limitation — the mode grows exactly as fast as the drive it leaves itself. Equivalently, by (7.3): the wave can at best grow as fast as the beams' color charges precess *in the beam frame*; faster growth would outrun its own energy supply.

Step 7.2.3 (Ceiling, derivation 2 (local drive branch)). Alternatively, the homogeneous charge-block branch (5.9) gives $\gamma^2 = G(w)$ pointwise; maximizing the implicit solution over w means solving $\gamma^2 = G(w^*) = \alpha V_0^2/\gamma$, i.e. $\gamma = (\alpha V_0^2)^{1/3}$ exactly. The ceiling is thus the maximum local growth rate of the drive branch, attained on the resonance surface; any global (quantized, windowed) mode sits below it. Measured dominant-branch peaks reach 95–99% of $(\alpha V_0^2)^{1/3}$ in every validated series.

7.3 Natural units and the universal well

Define, as in (1.11), $\Gamma = (\alpha V_0^2)^{1/3}$, $K = (\alpha/V_0)^{1/3}$, and hatted variables

$$\gamma = \Gamma \hat{\gamma}, \quad k_z = K \hat{k}, \quad w = K \hat{w}, \quad u = K \hat{u}. \quad (7.6)$$

Step 7.3.1 (Scale out the drive). Using $V_0^2 K^2 = V_0^2 (\alpha/V_0)^{2/3} = (\alpha V_0^2)^{2/3} = \Gamma^2$ and $\alpha V_0^3 K = \alpha^{4/3} V_0^{8/3} = \Gamma^4$:

$$G(w) = \frac{2\alpha V_0^3 K \hat{w}}{\Gamma^2 \hat{\gamma}^2 + V_0^2 K^2 \hat{w}^2} = \Gamma^2 \frac{2\hat{w}}{\hat{\gamma}^2 + \hat{w}^2} \equiv \Gamma^2 \hat{G}(\hat{w}). \quad (7.7)$$

The drive is universal in hatted variables: no α , V_0 , or ε_s remains in $\hat{G}$.

Step 7.3.2 (Universal well condition and edges). $G > \gamma^2$ becomes $\hat{G} > \hat{\gamma}^2$:

$$\boxed{\hat{\gamma}^2 (\hat{\gamma}^2 + \hat{w}^2) < 2\hat{w}} \quad (7.8)$$

The edges solve $\hat{\gamma}^2 \hat{w}^2 - 2\hat{w} + \hat{\gamma}^4 = 0$:

$$\hat{w}_{\pm} = \frac{1 \pm \sqrt{1 - \hat{\gamma}^6}}{\hat{\gamma}^2}, \quad (7.9)$$

real only for $\hat{\gamma} \leq 1$ — a third, independent derivation of the ceiling (7.5). As $\hat{\gamma} \rightarrow 1$ the well closes onto the resonance point $\hat{w} = 1$ (i.e. $w = K$, $\gamma/V_0 = K$); see Fig. 6, right.

Step 7.3.3 (Provable non-collapse in αV_0). Rates scale as $(\alpha V_0^2)^{1/3}$ and wavenumbers as $(\alpha/V_0)^{1/3}$: no combination of the two is a function of the product αV_0 alone. This proved that the survey-era attempt to collapse the data on αV_0 had to fail, and supplied the variables for the successful collapse test (T2.5: $\gamma_{\text{peak}}/\Gamma = 0.977 \pm 0.011$ across $\alpha = 1\text{--}3$, $V_0 = 0.03\text{--}0.1$ at fixed window group and $\varepsilon_s = 0.15$).

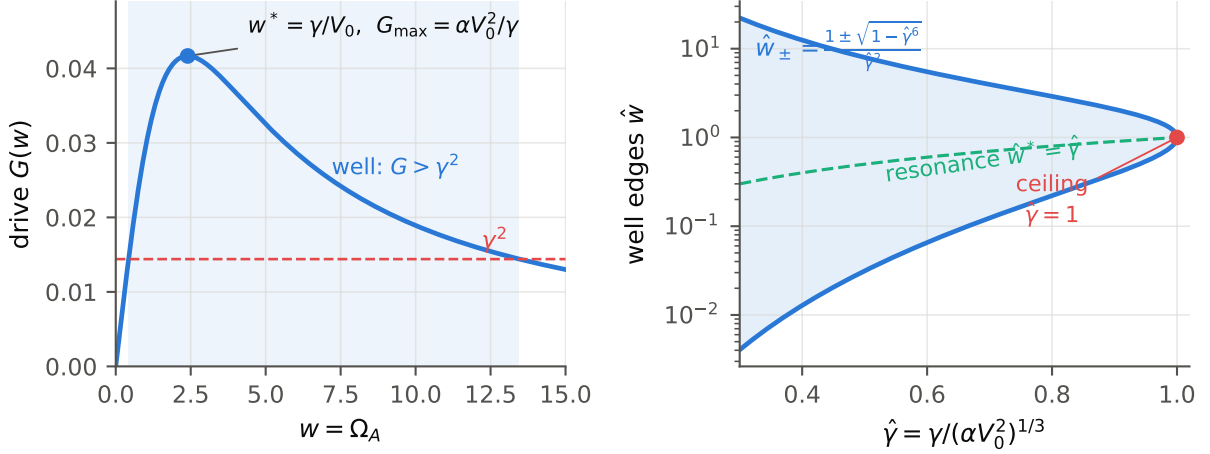


Figure 6: The precession-resonance drive. *Left*: $G(w)$ of (7.2) at $\alpha = 2$, $V_0 = 0.05$, $\gamma = 0.12$: maximal on the resonance surface $w^* = \gamma/V_0$ with $G_{\max} = \alpha V_0^2/\gamma$ (eq. (7.4)); a confining well exists wherever $G > \gamma^2$ (dashed), the shaded band $w \in (w_-, w_+)$. *Right*: the same in natural units, eq. (7.9): the well edges $\hat{w}_{\pm}$ close onto the resonance point as $\hat{\gamma} \rightarrow 1$ — the graphical statement of the ceiling $\gamma^3 \leq \alpha V_0^2$.

7.4 The resonance surface in space

Setting $w(\xi) = k_z - u(\xi) = w^* = \gamma/V_0$ and inverting (1.9):

$$\xi_{\text{res}} \simeq \frac{k_z - \gamma/V_0}{\alpha V_0} + \ln 2 \quad (7.10)$$

— the resonance surface moves *outward linearly with k_z* . The eigenmode peak tracks it (measured: peak ξ from -1 to -16 across $k_z = 0.5$ – 10 at the C25 parameters, -36 at cut 40), which is the geometric fact behind the confinement-set dispersion peak of §9.

8 Quantization of the shear branch: models A and B

The well is shallow: the branch holds *less than a quarter wavelength* of action at production parameters (measured $S = 0.31\pi$ – 0.46π), so naive Bohr–Sommerfeld $(n + \frac{1}{2})\pi$ quantization fails and the connection conditions at the two edges do all the work.

8.1 Geometry of the well

Fix the driven side ($\xi < 0$ for the $2 + i3$ pair). Moving inward from the wall:

- **Outer edge:** the window radius ξ_w (hard cut $\rightarrow$ zero-flux/Neumann face for a , §6.4); or, if the window is loose, the free turning point where G falls back to γ^2 (at $w = \hat{w}_-$ side, eq. (7.9)).
- **Inner edge:** the “knee” ξ_0 where the v^3 factor switches the drive off over $\lesssim 1$ ξ -unit. Since $g = -G |\tanh \xi|^3$ near the layer-side edge of the saturated region, the well ends where

$$G |\tanh \xi_0|^3 = \gamma^2 \quad \implies \quad \tanh^3 \xi_0 = \frac{\gamma^2}{G}, \quad (8.1)$$

giving $\xi_0 \approx 1.4$ – 2 at production parameters (the code’s model A uses the fixed representative $\xi_0 = 1.5$).

- **Interior:** $Q > 0$ from (6.10). In the window-limited regime relevant near the dispersion peak, $u \ll k_z$ across the well, so $\mathcal{D} \approx \gamma^2 + k_z^2$ and $G \approx G(k_z)$ vary by $< 5\%$ across the well — the numerically confirmed *square-well regime*.
- **Beyond the inner edge** ($|\xi| < \xi_0$): $g \rightarrow 0$, so $Q \rightarrow -\mathcal{D} \approx -(\gamma^2 + k_z^2)$: evanescent with decay constant

$$\kappa = \sqrt{\gamma^2 + k_z^2} \approx k_z. \quad (8.2)$$

8.2 The sharp-edge connection formula

Step 8.2.1 (Interior solution). Let $k_x = \sqrt{Q}$ (constant in the square-well regime) and measure distance s inward from the outer wall. The Neumann face ($a' = 0$ at $s = 0$) selects

$$a(s) = \cos(k_x s). \quad (8.3)$$

Step 8.2.2 (Log-derivative at the inner edge). At the inner edge $s = \Delta$ (well width), with total action $S = k_x \Delta$:

$$\left. \frac{a'}{a} \right|_{s=\Delta^-} = -k_x \tan S. \quad (8.4)$$

Step 8.2.3 (Evanescent side). For $s > \Delta$ the solution decaying toward the layer is $a \propto e^{-\kappa(s-\Delta)}$, with $a'/a = -\kappa$. The drive switches off over a distance short compared with $2\pi/k_x$ (the “sharp step”), so a and a' are continuous across it:

$$-k_x \tan S = -\kappa \quad \implies \quad \boxed{\tan S = \frac{\kappa}{k_{\text{edge}}}} \quad (k_{\text{edge}} = k_x \text{ at the inner edge}), \quad (8.5)$$

with $S \in (0, \pi/2)$ for the nodeless $n = 0$ mode and $S \in (n\pi, n\pi + \pi/2)$ for the n -th overtone. Because $\tan S > 0$ requires $S < \pi/2 \pmod{\pi}$, the branch is *quantization-marginal*: it holds less than a quarter wave, and γ adjusts to sit close below the ceiling so that the well is exactly this shallow. Both the matching geometry and the graphical solution of model A’s closure are shown in Fig. 7.

Physical picture. A quarter-wave well. The condition is not “fit $n + \frac{1}{2}$ waves in the well” but “match one shallow cosine arc to its evanescent tail”: a surface-mode-like state. This marginality explains two robust observations at once. First, why measured rates hug the ceiling: lowering γ deepens the well quickly (both G and Q rise), so only a small drop below the ceiling buys all the action the connection condition demands. Second, why rates are insensitive to window details (few percent) while the *peak position* is set entirely by them: γ is pinned near the ceiling wherever a well exists at all, and the geometry only decides for which k_z that is possible (§9).

8.3 Model A: closed-form dispersion relation

Model A adopts the square-well regime literally and closes everything analytically.

Step 8.3.1 (Assumptions). (i) $u \ll k_z$ over the well, so $w \approx k_z$ and the drive is the *Doppler-detuned* value

$$G = G(k_z) = \frac{2\alpha V_0^3 k_z}{\gamma^2 + V_0^2 k_z^2}; \quad (8.6)$$

(ii) $\mathcal{D} \approx \gamma^2 + k_z^2$ constant; (iii) well width $\Delta x = \varepsilon_s (\xi_w - \xi_0)$ with $\xi_0 = 1.5$.

Step 8.3.2 (Interior wavenumber). From (6.10) with these approximations,

$$k_x = \sqrt{Q} = \sqrt{\frac{(\gamma^2 + k_z^2)(G - \gamma^2)}{\gamma^2}} = \sqrt{\gamma^2 + k_z^2} \frac{\sqrt{G - \gamma^2}}{\gamma} \approx k_z X, \quad \boxed{X \equiv \frac{\sqrt{G - \gamma^2}}{\gamma}} \quad (8.7)$$

(using $k_z \gg \gamma$ in the prefactor, consistent with (i)).

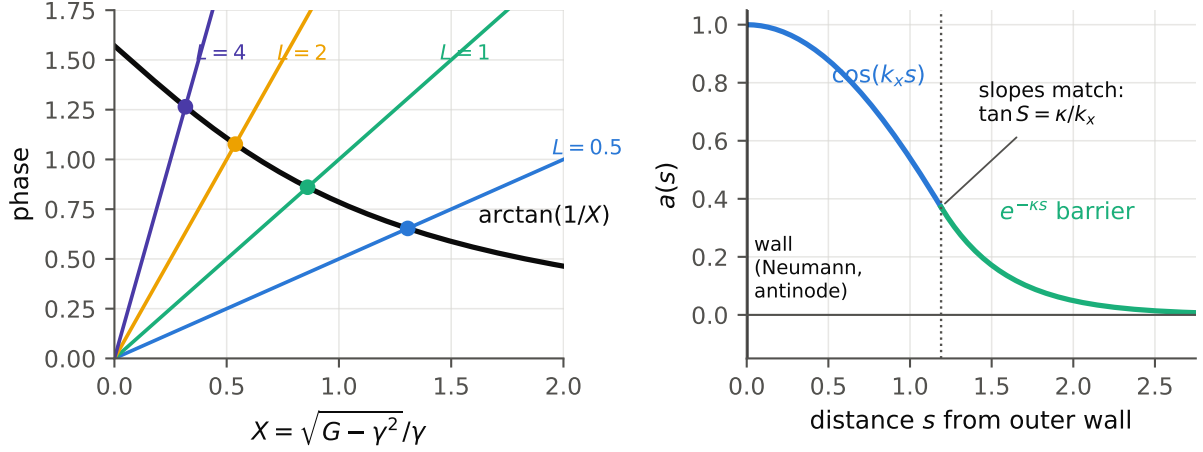


Figure 7: Quantization mechanics. *Left*: graphical solution of model A's closure $LX = \arctan(1/X)$, eq. (8.8) — the geometry factor $L = \varepsilon_s(\xi_w - \xi_0)k_z$ alone fixes the interior-wavenumber ratio X , with no reference to γ . *Right*: the sharp-edge matching behind (8.5) (drawn for $k_x = 1$, $\kappa = 2.5$): a cosine arc with its antinode at the Neumann wall meets the evanescent barrier tail with continuous value and slope, which happens exactly when $\tan S = \kappa/k_x$; the arc is always shorter than a quarter wave.

Step 8.3.3 (Quantization closes on X alone). Insert $k_x = k_z X$ and $\kappa \approx k_z$ into (8.5):

$$S = k_x \Delta x = \underbrace{\varepsilon_s(\xi_w - \xi_0)k_z}_{\equiv L} X, \quad \tan S = \frac{k_z}{k_x X} = \frac{1}{X} \implies \boxed{LX = \arctan \frac{1}{X}} \quad (8.8)$$

— a single transcendental equation for X that contains *no* γ : the geometry factor L (window, shear width, wavenumber) fixes X by itself. X decreases monotonically as L grows (looser window $\Rightarrow$ smaller interior wavenumber needed).

Step 8.3.4 (Recover γ). By the definition of X , $G = \gamma^2(1 + X^2)$; equate to (8.6):

$$\gamma^2(1 + X^2)(\gamma^2 + V_0^2 k_z^2) = 2\alpha V_0^3 k_z. \quad (8.9)$$

This is a quadratic in γ^2 :

$$\gamma^4 + V_0^2 k_z^2 \gamma^2 - \frac{2\alpha V_0^3 k_z}{1 + X^2} = 0, \quad (8.10)$$

whose positive root is

$$\gamma^2 = \frac{1}{2} \left[-V_0^2 k_z^2 + \sqrt{V_0^4 k_z^4 + \frac{8\alpha V_0^3 k_z}{1 + X^2}} \right] \quad (8.11)$$

— model A. Validated within $\sim 3\%$ of the σ -chased dominant eigenvalues near and above the dispersion peak, and it identifies the peak within ± 0.5 in k_z .

8.4 Model B: exact-action quantization

Model B keeps the exact $Q(x; \gamma)$ of (6.10) — no square-well or $w \approx k_z$ approximation — and root-finds the quantization phase. For trial γ :

1. compute Q on a fine grid of the driven side; find the largest contiguous region $Q > 0$ (the well) inside the window radius ξ_w ;

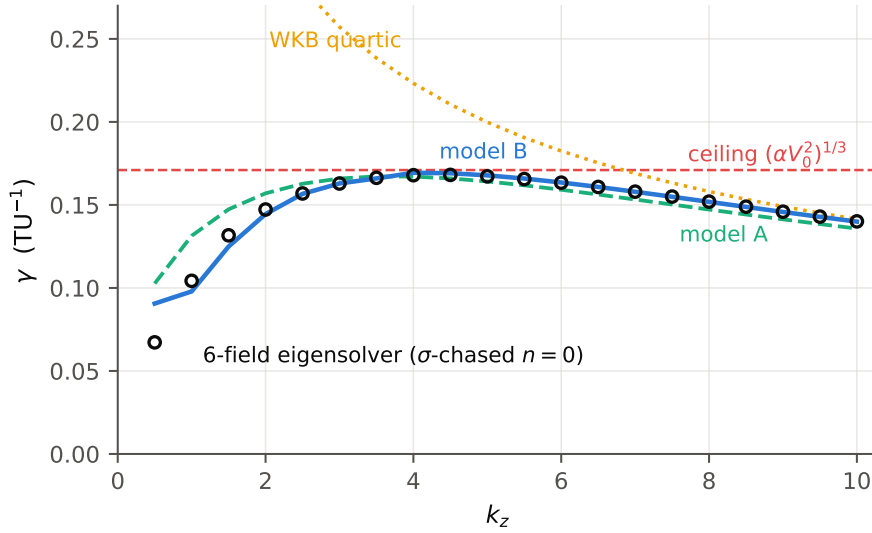


Figure 8: The full theory chain on one dispersion curve ($\alpha = 2$, $V_0 = 0.05$, $\varepsilon_s = 0.15$, hard cut $\xi_w = 11$; regenerated for this document, no cached values). Open circles: dominant (σ -chased $n = 0$) eigenvalues of the six-field system (4.27) at $N_x = 384$. Solid: model B (§8.4) — agreement $\leq 0.2\%$ for $k_z \geq 2.5$ here, degrading to a few percent only at $k_z \lesssim 1.5$ where the well is least square. Dashed: closed-form model A (8.11). Dotted: the WKB quartic (2.4), overshooting at low k_z exactly as diagnosed in §11. The curve peaks at $k_z = 4.5$ at 98% of the ceiling (dashed red), and the peak location matches eq. (9.2): $0.1 \times (11 - \ln 2) + 3.42 = 4.45$.

2. exact action $S(\gamma) = \int_{\text{well}} \sqrt{Q} dx$;
3. edge wavenumber $k_{\text{edge}} = \sqrt{Q}$ evaluated just inside the *inner* edge; $\kappa = \sqrt{\gamma^2 + k_z^2}$;
4. connection phase Φ :

$$\Phi = \begin{cases} \arctan(\kappa/k_{\text{edge}}) & \text{well attached to the tanh}^3 \text{ knee (sharp inner step),} \\ \pi/4 & \text{well detached (soft inner turning point),} \end{cases} \quad (8.12)$$

with an additional $+\pi/4$ if the outer edge is a free turning point rather than the wall, plus $n\pi$ for overtones. The $\pi/4$ terms are the standard Airy connection phases at first-order turning points (the linear-turning-point matched asymptotics that replace the sharp-step matching of §8.2); the Neumann wall contributes phase 0 (cosine antinode at the wall, exactly as in (8.5)).

5. solve $S(\gamma) = \Phi(\gamma)$, scanning γ downward from just below the ceiling so the first bracket found is $n = 0$.

Validation: across 12 series spanning $\alpha = 1\text{--}5$, $V_0 = 0.03\text{--}0.2$, windows $\xi_w = 5\text{--}40$, $k_z = 0.5\text{--}10$, model B matches the σ -chased dominant ($n = 0$) six-field eigenvalues to $\leq 2\%$ (median $\leq 0.2\%$) at every $k_z \geq 1.5$; the worst deviations (6–9%) occur at $k_z \lesssim 1$ where the well is widest and least square. Figure 8 shows this comparison, regenerated from scratch for this document at the reference production point, together with the quartic it supersedes.

8.5 Overtone ladder and eigenvalue-selection hygiene

With $S(\gamma)$ decreasing in γ , the roots of $S(\gamma) = \Phi + n\pi$ form a ladder $\gamma_0 > \gamma_1 > \dots$ with spacing

$$\Delta\gamma \simeq \frac{\pi}{|\partial S/\partial \gamma|}, \quad (8.13)$$

a few $\times 10^{-2} \text{ TU}^{-1}$ at loose windows. A shift-invert eigensolver returns the eigenvalue nearest its target σ ; the legacy default $\sigma = 0.55 \gamma_{\text{WKB}}(\text{quartic})$ lands mid-ladder wherever the quartic underestimates (low α , or k_z above the quartic's peak), silently selecting an overtone. Consequence (referee-critical, discovered via this theory): at $(\alpha, V_0, k_z) = (1, 0.05, 4)$, window 20, the dominant mode has $\gamma = 0.134$ but default- σ returned 0.106 and the cached survey curve carried 0.082 — an $n \approx 3$ overtone; the eigenmode-seeded simulation then faithfully grew the overtone it was seeded with (plateau-audited 0.081). *Simulation-vs-solver agreement certifies numerics, not dominant-mode selection.* All reference data now use σ -chasing (re-shifting σ upward until the found window is clear).

9 The dispersion curve: confinement-set peak and windowed roll-off

9.1 No intrinsic k_z maximum

In an unbounded domain nothing stops the eigenmode from following the resonance surface (7.10) outward as k_z grows: the drive at resonance is always $G_{\text{max}} = \alpha V_0^2 / \gamma$, independent of k_z , so $\gamma(k_z)$ rises monotonically and saturates at the ceiling (7.5) while the mode migrates outward. Directly demonstrated (eigensolver, $\alpha = 2$, $V_0 = 0.05$, cut 40): γ climbs to 0.9955Γ by $k_z = 10$ with no turnover, mode peak at $\xi = -36$. The non-monotonic measured dispersion is therefore *windowed physics*, not an intrinsic wavelength selection.

9.2 The peak: resonance surface meets the window

Step 9.2.1. For k_z below the peak, the resonance surface $\xi_{\text{res}}(k_z)$ of (7.10) lies inside the window and the mode enjoys the full resonant drive: γ near the ceiling.

Step 9.2.2. Once ξ_{res} reaches the window radius ξ_w , the well is truncated at the wall value of the covariant wavenumber, $u_w = \alpha V_0(\xi_w - \ln 2)$, the resonance is no longer accessible, and only the detuned drive $G(k_z)$ of (8.6) remains: γ rolls off (next subsection). The peak is the crossover $\xi_{\text{res}}(k_z) = \xi_w$:

$$\frac{k_z - \gamma/V_0}{\alpha V_0} + \ln 2 = \xi_w. \quad (9.1)$$

Step 9.2.3. At the crossover the mode still runs at essentially the ceiling, so substitute $\gamma = \Gamma = (\alpha V_0^2)^{1/3}$, i.e. $\gamma/V_0 = (\alpha/V_0)^{1/3} = K$:

$$k_z^{\text{peak}} \simeq \alpha V_0 (\xi_w - \ln 2) + c \left(\frac{\alpha}{V_0} \right)^{1/3}, \quad c = 1.0 \pm 0.1 \quad (9.2)$$

(c absorbs the residual quantization offset; the empirical fit across 11 series gives c in 0.85–1.0). Both terms are *independent of ε_s* : the shear width does not select the wavelength — the coupling and the containment radius select it together. Verified to ± 0.6 in k_z on all 11 usable series, including the direct window test: moving the cut $11 \rightarrow 25$ at fixed physics ($\alpha = 2$, $V_0 = 0.05$) moved the measured peak $4.5 \rightarrow 5.5$ (formula: $4.45 \rightarrow 5.85$). Both tests are shown in Fig. 9.

Physical picture. The fence picks the wavelength. Every k_z wants to sit on its own resonance shell $\xi_{\text{res}}(k_z)$, and the shells march outward linearly with k_z . The window is a fence: modes whose shell is inside run at essentially the ceiling; the first shell outside the fence marks the peak; beyond it the mode is pinned at the wall with only the detuned drive left, and rolls off. Nothing about the layer width ε_s enters — for a Kelvin–Helmholtz-type instability this is the most counter-intuitive single statement of the theory, and it is what the data (peak drifting with window radius at fixed physics) confirmed directly.

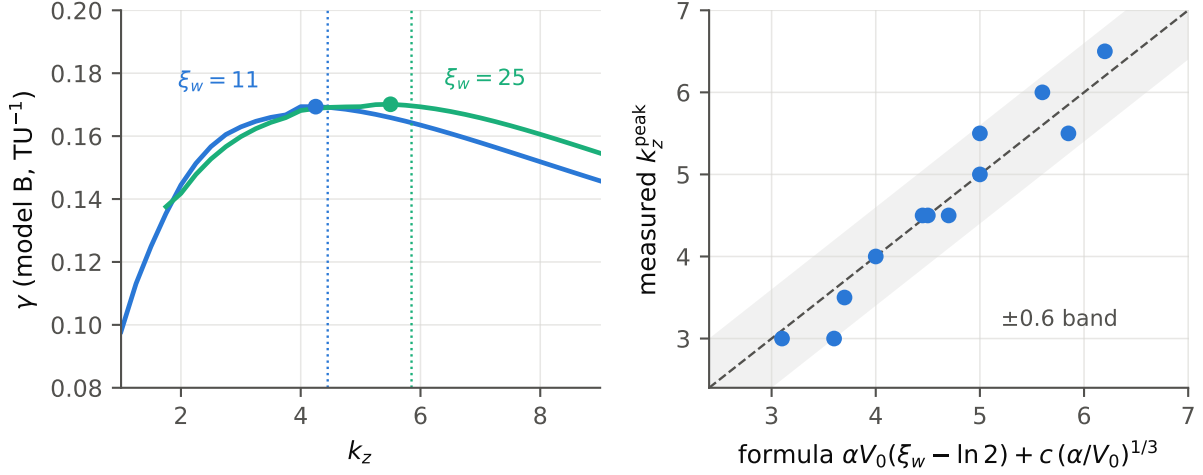


Figure 9: The confinement-set peak. *Left*: model-B dispersion curves at fixed physics ($\alpha = 2$, $V_0 = 0.05$, $\varepsilon_s = 0.15$) for two window radii: widening the cut $11 \rightarrow 25$ moves the peak (dots) from $k_z \approx 4.3$ to ≈ 5.6 ; dotted verticals are formula (9.2) with $c = 1$. *Right*: measured dominant-branch peaks vs the formula across the validation series ($\alpha = 1\text{--}5$, $V_0 = 0.03\text{--}0.2$, $\xi_w = 5\text{--}25$; the shaded band is ± 0.6 in k_z).

9.3 Beyond-peak roll-off

Past the peak, model A applies with the window-truncated well. Two limits of (8.11):

Step 9.3.1 (Loose-window limit $X \rightarrow 0$). $1 + X^2 \rightarrow 1$ gives the envelope of the truncated branch,

$$\gamma^2 = \frac{1}{2} \left[-V_0^2 k_z^2 + \sqrt{V_0^4 k_z^4 + 8\alpha V_0^3 k_z} \right]. \quad (9.3)$$

Step 9.3.2 (High- k_z asymptotics). For $8\alpha/(V_0 k_z^3) \ll 1$ expand the square root, $\sqrt{1 + \epsilon} \simeq 1 + \epsilon/2$ with $\epsilon = 8\alpha/(V_0 k_z^3)$:

$$\gamma^2 \simeq \frac{1}{2} V_0^2 k_z^2 \cdot \frac{4\alpha}{V_0 k_z^3} = \frac{2\alpha V_0}{k_z} \quad \Rightarrow \quad \boxed{\gamma \simeq \sqrt{\frac{2\alpha V_0}{k_z}}} \quad (9.4)$$

— the measured beyond-peak decline ($\gamma \propto k_z^{-1/2}$), steepened further by the $(1 + X^2)^{-1}$ quantization factor as the truncated well narrows.

9.4 Why the data said $k_z^{\text{peak}} \approx 2\alpha$; γ^{peak}

Over the surveyed grid ($\alpha = 1\text{--}5$, $V_0 = 0.03\text{--}0.1$, window radii shrinking as α grew, roughly $\alpha V_0 \xi_w \approx 1$), formula (9.2) numerically tracks $\approx 2\alpha$: the sub-linear resonance term $(\alpha/V_0)^{1/3}$ plus a window term whose radius the campaign design shrank as α grew mimics a linear trend across a factor-5 range of α . The “ $k_z^{\text{peak}} \approx 2\alpha$ law” is a serviceable survey-box mnemonic; (9.2) is the law. Similarly, the empirical $\gamma^{\text{peak}} \sim V_0^{0.85\text{--}0.92}$ at fixed α is the ceiling’s $V_0^{2/3}$ times the window/quantization correction $\hat{\gamma}(\text{budget}, \alpha V_0 \xi_w)$, which increases with V_0 across the surveyed range; $\gamma^{\text{peak}} = (0.95\text{--}0.99) \Gamma$ for $\xi_w \geq 8$, dropping to $\sim 0.95 \Gamma$ at $\xi_w = 5$.

10 Shear-width scaling: the quantization budget

How does ε_s enter, given that neither the ceiling (7.5) nor the peak formula (9.2) contains it? Answer: only through the *action*, i.e. the quantization budget. We nondimensionalize the exact

action integral step by step. (This underlies the EPS-scan v2 campaign design and its Phase-0 theory, `analysis/eps_collapse_theory.py`.)

Step 10.0.1 (Exact action in the w variable). In the saturated region, $u = \alpha V_0(|\xi| - \ln 2)$ and $x = \varepsilon_s \xi$ give $dx = \varepsilon_s d\xi = \frac{\varepsilon_s}{\alpha V_0} du$, and $w = k_z - u$ makes $|dw| = |du|$:

$$S = \int_{\text{well}} \sqrt{Q} dx = \frac{\varepsilon_s}{\alpha V_0} \int_{\text{well}} \sqrt{Q} dw. \quad (10.1)$$

Step 10.0.2 (Hatted integrand). With $w = K\hat{w}$ and (7.7),

$$\mathcal{D} = \gamma^2 + \Omega_A \Omega_F = \gamma^2 + w(2k_z - w) = K^2 \left[\hat{w}(2\hat{k} - \hat{w}) + V_0^2 \hat{\gamma}^2 \right] \equiv K^2 \hat{\mathcal{D}}, \quad (10.2)$$

$$Q = \frac{\mathcal{D}(G - \gamma^2)}{\gamma^2} = K^2 \frac{\hat{\mathcal{D}}(\hat{G} - \hat{\gamma}^2)}{\hat{\gamma}^2} \equiv K^2 \hat{Q}(\hat{w}; \hat{k}, \hat{\gamma}) \quad (10.3)$$

(the factors Γ^2 from $G - \gamma^2$ and γ^2 cancel; the explicit $V_0^2 \hat{\gamma}^2 \leq 0.04$ inside $\hat{\mathcal{D}}$ is negligible over the production range, so $\hat{Q}$ is universal in hatted variables).

Step 10.0.3 (Assembled scaling). Using $dw = K d\hat{w}$ together with

$$\frac{K^2}{\alpha V_0} = \left(\frac{\alpha}{V_0} \right)^{2/3} \frac{1}{\alpha V_0} = \alpha^{-1/3} V_0^{-5/3} = \frac{1}{V_0 (\alpha V_0^2)^{1/3}}, \quad (10.4)$$

we obtain

$$S = \frac{\varepsilon_s K^2}{\alpha V_0} \int \sqrt{\hat{Q}} d\hat{w} = \frac{\varepsilon_s}{V_0 (\alpha V_0^2)^{1/3}} \hat{I}(\hat{k}, \hat{\gamma}), \quad \hat{I} \equiv \int_{\hat{w}\text{-well}} \sqrt{\hat{Q}} d\hat{w} \quad (10.5)$$

(numerically verified: at $\alpha = 2$, $V_0 = 0.05$, $\varepsilon_s = 0.15$, $k_z = 4$, $\gamma = 0.16$, both sides give $S \approx 2.8$ before quantization pulls γ up). The shear width therefore enters the eigenvalue problem *only* as a multiplicative budget on the action; the well's shape in $\hat{w}$ is ε_s -free.

Remark 10.1 (Normalization of the program's “ P ” group). The project's shorthand (PHYSICS.md eq. (11.5)) writes $S = (\varepsilon_s/\Gamma) \int \sqrt{\hat{Q}'} d\hat{w}$ with the group $P = \Gamma/\varepsilon_s$; comparing with (10.5), that convention absorbs a factor V_0^{-2} into its $\hat{Q}'$ ($\hat{Q}' = \hat{Q}/V_0^2$). The clean statement is (10.5): the dimensionless budget prefactor is $\varepsilon_s^{-1} V_0 (\alpha V_0^2)^{1/3} = (\alpha V_0^5)^{1/3} / \varepsilon_s$, with the integral $\hat{I}$ universal. Either bookkeeping gives identical model-A/B numbers (the campaign tooling evaluates the models directly and never uses the shorthand); the distinction matters only if one collapses *data* against a single “budget group”, where the residual V_0 -dependence is part of why a pure one-group collapse in $P = (\alpha V_0^2)^{1/3} / \varepsilon_s$ is imperfect (cf. the observed ε_s -drift of k_z^{peak} at fixed window).

Remark 10.2 (Qualitative consequences). Larger ε_s (at fixed window and physics) enlarges S at given γ , and since S decreases in γ , quantization then pushes γ *up* toward the ceiling — the sign of the measured 10–39% rise of γ from $\varepsilon_s = 0.10 \rightarrow 0.45$ at fixed $(k_z, \xi_{\text{sponge}})$; Fig. 10 shows the model-B version of this rise. Because the budget also shifts which k_z first saturates its quantization, k_z^{peak} acquires a mild ε_s -dependence even though (9.2)'s leading terms have none.

Physical picture. ε_s buys action. Stretching the layer stretches the well in physical x while leaving its depth and shape unchanged in the scaled variable $\hat{w}$ — so at fixed γ the mode accumulates more phase, and the connection condition then lets γ float up toward the ceiling. The shear width is a *budget* knob (how much quantization tax the mode pays), not a wavelength selector; that is why its effect shows up as a fractional rise of γ everywhere rather than as a shift of the physics scales Γ , K .

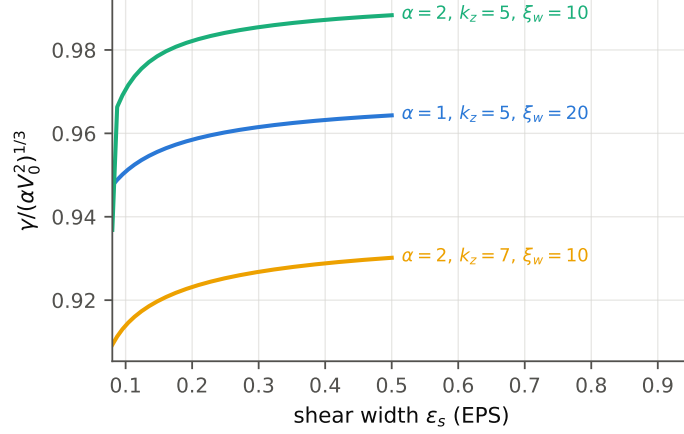


Figure 10: The quantization budget at work (model B, beyond-peak window-limited points): at fixed physics and window, increasing the shear width ϵ_s raises γ toward the ceiling, exactly as eq. (10.5) prescribes — ϵ_s enters only as a multiplier on the action.

11 Diagnosis of the WKB quartic’s domain of validity

The quartic (2.4) models the drive as the constant $-\alpha^2 V_0 k_z / \omega^2$ term of (1.3) — one power of v (from the current), one resonance-free factor $\alpha^2 V_0 k_z$. The true drive (7.2) has three powers of v , is Doppler-resonant, and is ceiling-bounded by (7.4). Comparing the two fixes exactly where eq. (1.4)/(2.4) can and cannot work:

- **Low k_z (below the true peak):** the quartic’s drive $\alpha^2 V_0 k_z / \gamma^2$ has no ceiling; as γ is iterated it can supply arbitrarily large growth, and the quartic overshoots the true branch by the measured 10–20 $\times$ in the $k_z \lesssim 1$ tail.
- **High k_z :** the quartic’s balance $\omega^4 - k_z^2 \omega^2 \approx \alpha^2 V_0 k_z$ makes $\gamma^2 \approx \alpha^2 V_0 / k_z$, a *decline*, while the true unbounded branch saturates at the ceiling; in windowed data the truncated true branch declines as (9.4) instead. In the weak-coupling corner this produces the observed $\sim 25\%$ quartic underestimates.
- **$k_z = 0$:** the geometry changes (parabolic potential exact, different channel); the harmonic-core quantization is exact there and the 0.5% chromo-Weibel anchor (2.8) stands untouched by this critique.

The quartic is thus a valid *harmonic-core* theory (its own regime: $k_z = 0$ and the immediate neighborhood of a parabolic well) but not a theory of the finite- k_z shear branch; models A/B replace it there.

12 Cold two-stream rates in code units

The counter-streaming beams support, away from the shear layer ($v_z \rightarrow \pm V_0$), an electrostatic color-1 two-stream instability. Its exact cold rates in code units calibrate the bandpass-filter design (Appendix B) and were used to test (and reject) the outer two-stream as the late-catastrophe driver.

12.1 Dispersion relation

Step 12.1.1. Longitudinal channel: perturbations $\propto e^{i(kz - \omega t)}$ of n_s , v_{zs} , coupled by the longitudinal field E_z . Linearized continuity and momentum for beam s with background velocity u_s (here

$u_{A,B} = \pm V$), per-beam plasma frequency (1.5):

$$\delta n_s = \frac{k n_0 \delta v_s}{\omega - k u_s}, \quad \delta v_s = \frac{i q_s E_z / m}{-(i)(\omega - k u_s)} \implies \delta n_s = -\frac{n_0 q_s E_z k / m}{(\omega - k u_s)^2} \cdot \frac{1}{i} \cdot i = -\frac{k q_s n_0 E_z / m}{(\omega - k u_s)^2}. \quad (12.1)$$

Step 12.1.2. Poisson/Gauss ($\varepsilon_0 = 1$): $ikE_z = \sum_s q_s \delta n_s$ gives

$$ikE_z = -\sum_s \frac{k q_s^2 n_0 E_z / m}{(\omega - k u_s)^2} \cdot \frac{1}{i^{-1}} \implies 1 = \sum_s \frac{\omega_p^2}{(\omega - k u_s)^2} = \frac{1}{(\omega - kV)^2} + \frac{1}{(\omega + kV)^2}. \quad (12.2)$$

(the standard two-beam form; each beam contributes $\omega_p^2 = 1$).

12.2 Closed-form growth rate

Step 12.2.1 (Reduce to a quadratic). Put the right side of (12.2) over a common denominator, using $(\omega \mp kV)^2$ sum = $2(\omega^2 + k^2 V^2)$:

$$(\omega^2 - k^2 V^2)^2 = 2(\omega^2 + k^2 V^2). \quad (12.3)$$

Let $y = \omega^2$ and $\beta = k^2 V^2$:

$$(y - \beta)^2 = 2(y + \beta) \iff y^2 - 2(\beta + 1)y + \beta^2 - 2\beta = 0. \quad (12.4)$$

Step 12.2.2 (Solve).

$$y_{\pm} = (\beta + 1) \pm \sqrt{(\beta + 1)^2 - \beta^2 + 2\beta} = (\beta + 1) \pm \sqrt{4\beta + 1}. \quad (12.5)$$

Step 12.2.3 (Instability condition). $y_- < 0$ (i.e. $\omega^2 < 0$, purely growing) iff $(\beta + 1)^2 < 4\beta + 1$ iff $\beta^2 < 2\beta$ iff $\beta < 2$:

$$\boxed{kV < \sqrt{2}} \quad \text{unstable band.} \quad (12.6)$$

At $V_0 = 0.1$ this is $k < 14.1$ — exactly the documented $k_{ts} \approx 14$ two-stream band that fixed the production bandpass edge.

Step 12.2.4 (Growth rate, maximum, small- k limit).

$$\boxed{\gamma^2(kV) = -y_- = \sqrt{4(kV)^2 + 1} - (kV)^2 - 1} \quad (12.7)$$

Maximize over β : $d\gamma^2/d\beta = 2/\sqrt{4\beta + 1} - 1 = 0$ gives $\beta = 3/4$, i.e.

$$kV = \frac{\sqrt{3}}{2}, \quad \gamma_{\max}^2 = \sqrt{4} - \frac{3}{4} - 1 = \frac{1}{4}, \quad \gamma_{\max} = \frac{1}{2}. \quad (12.8)$$

Small kV : $\sqrt{4\beta + 1} \simeq 1 + 2\beta - 2\beta^2$ gives $\gamma^2 \simeq \beta$, i.e. $\gamma \simeq kV$. (Measured full-domain rates 0.7–0.9 exceed $\frac{1}{2}$ because they include the electromagnetic/Weibel corrections outside this electrostatic model.) The full curve is Fig. 11.

Physical picture. Bunching in a moving mirror. In the frame of one beam, the other beam's charge bunches stream past and their electric field does work on the first beam's particles, bunching them in turn; below the band edge the bunches of each beam arrive in phase with the wells of the other and the loop closes. The band edge $kV = \sqrt{2}$ is where the streaming Doppler shift finally outruns the plasma response. In code units ($\omega_p = 1$ per beam) the band at $V_0 = 0.1$ ends at $k_z = 14.1$ — the number that fixed the production bandpass-filter edge, and the theory whose *spatial* predictions (growth localized in the outer $|v_z| \approx V_0$ region) were used to rule the two-stream out as the late-catastrophe driver.

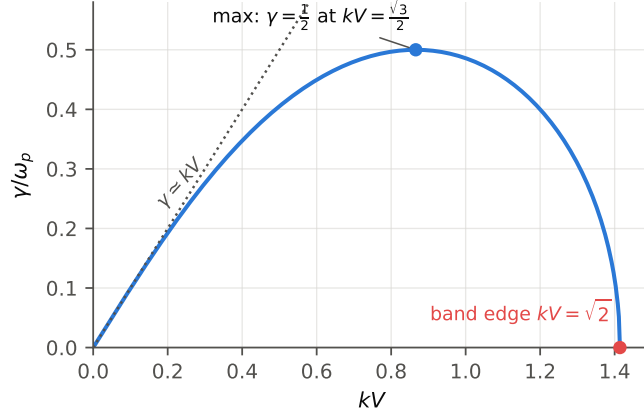


Figure 11: Exact cold two-stream growth rate (12.7) in code units: $\gamma \simeq kV$ at long wavelength, maximum $\gamma = \frac{1}{2}$ at $kV = \sqrt{3}/2$, band edge at $kV = \sqrt{2}$ (eq. (12.6)).

13 Nonlinear theory: cavitation threshold and the back-reaction law

13.1 Density cavitation of the finite-amplitude KH mode

The linear eigensolver state vector contains no fluid density/momentum blocks; the late-time “hard wall” at $V_0 \gtrsim 0.08$ is a nonlinear fluid effect. Its threshold is derived as follows.

Step 13.1.1 (First-order compressive response). Linearize (3.1) about $n = 1$ for a mode $\propto e^{\gamma t}$:

$$\gamma \delta n = -(\mathrm{i}k_z \delta p_z + \partial_x \delta p_x) \quad \implies \quad |\delta n| \sim \frac{k_z |p_{z1}| + k_x^{\mathrm{eff}} |p_{x1}|}{\gamma}, \quad (13.1)$$

where k_x^{eff} is the mode’s inverse x -scale.

Step 13.1.2 (Second-order (ponderomotive) part). Quadratic beats of the mode grow at 2γ and accumulate a z -mean density modification; the observed density-perturbation growth at the retained k_z at twice the saturating field rate is this quadratic part taking over as the fields saturate.

Step 13.1.3 (Cavitation threshold). Cavitation ($\delta n \rightarrow n_0$, i.e. n hits the code floor 0.05) occurs when the momentum amplitude reaches

$$p_{\mathrm{cav}} \sim \frac{\gamma}{\max(k_z, k_x^{\mathrm{eff}})} \quad (13.2)$$

(order of magnitude). Beyond it, the floored divisions $v = p/\max(n, 0.05)$ decouple momentum from density and the model no longer represents the physical system; a long-lived floored-pocket state persists for tens of TU before terminal blow-up. Because (13.2) is an *amplitude* threshold, whether a run cavitates within its length is controlled by drive strength (V_0 , through γ and the saturation level) and effective seed — hence the observed gradual $V_0 = 0.05 \rightarrow 0.10$ transition, its insensitivity to window mechanism and radius, and the practical rule of capping `target_tu` just past the plateau at high V_0 . Forensics on a failing run localized every growing channel at the shear layer (99.9% of density-perturbation power at $|\xi| < 5$) with the outer region quiet, ruling out the outer branches as the driver.

Physical picture. The mode digs its own grave. A growing wave in a compressible fluid carries a density modulation proportional to its momentum amplitude over γ ; on top of it, the quadratic (ponderomotive) beat pushes plasma out of the high-field regions at rate 2γ . When the excavated fraction reaches the background density, the fluid description — and the code’s floored version of it —

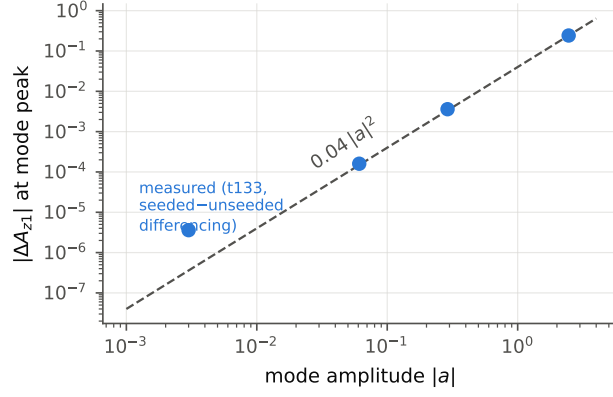


Figure 12: The measured back-reaction law against the derived quadratic structure: $|\Delta A_{z1}|$ at the mode peak vs the instantaneous mode amplitude $|a|$, from seeded-minus-unseeded differencing of unfrozen runs (t133; two configurations pooled). The dashed line is $0.04|a|^2$: slope 2 over four decades, endpoint exact (-0.242 vs -0.242 at $|a| = 2.46$). The *sign* deepens the background — no depletion.

stops representing the plasma. The threshold (13.2) is an *amplitude* cliff, not a time cliff: that single fact explains why the failure zone maps onto drive strength (V_0), why it cares little about window design, and why the only safe practice at high V_0 is to stop measuring once the linear plateau is banked.

13.2 Structure of the back-reaction on A_{z1} : the quadratic law

Unfreezing A_z^1 lets the color-2/3 wave imprint on the background. The measured law (two independent configurations, four decades of $|a|^2$, endpoint exact) is

$$\Delta A_{z1} \simeq -0.04|a|^2, \quad (13.3)$$

localized at the mode peak, i.e. the wave *deepens* $|A_{z1}|$ — opposite in sign to depletion. The structure (quadratic, $k_z = 0$, localized, growing at 2γ) is derived as follows; the coefficient and sign are taken from measurement.

Step 13.2.1 (All color-1 second-order sources are helicity beats). For real fields with circular structure $X^2 = \text{Re}(\hat{X}e^{ik_z z})$, $X^3 = \text{Im}(\hat{X}e^{ik_z z})$, any color-1 cross-product source is a beat of two color-2/3 waves, e.g. for the precession of Q^1 :

$$(\vec{Q} \times \vec{A}_z)^1 = Q^2 A_z^3 - Q^3 A_z^2 = \text{Re}(\hat{q}e^{i\theta}) \text{Im}(\hat{a}e^{i\theta}) - \text{Im}(\hat{q}e^{i\theta}) \text{Re}(\hat{a}e^{i\theta}) = \text{Im}(\hat{a} \hat{q}^*), \quad \theta = k_z z, \quad (13.4)$$

which is *independent of z* : the color-1 imprint of a circular mode is purely $k_z = 0$, and $\propto |\text{amplitudes}|^2 e^{2\gamma t}$. The same computation applies to the current beats ($\delta n \delta v_z$, $\delta Q^{2,3} \delta v_z$ contributions to J_z^1) and to the field commutator beats in (3.5)/(3.7).

Step 13.2.2 (Chain to ΔA_{z1}). The $k_z = 0$ color-1 source drives $J_z^{1(2)} \rightarrow E_z^1$ (Ampère- z), and $\partial_t A_{z1} = -E_z^1$ integrates it. With every second-order source $\propto e^{2\gamma t}$, each time integration divides by 2γ , so

$$\Delta A_{z1}(t) = -\frac{(\text{source strength})}{(2\gamma)^2} e^{2\gamma t} \propto -|a(t)|^2, \quad (13.5)$$

a pure quadratic law in the instantaneous mode amplitude, pinned at the mode peak — exactly the measured structure (13.3) (Fig. 12).

Physical picture. Waves weigh on their background. Any quadratic beat of a single-helicity mode is z -independent (eq. (13.4)) — a DC force growing at 2γ — so the wave exerts a steady, locally-confined push on the color-1 sector wherever its envelope peaks, like radiation pressure. Here the push happens to point “downhill”: it makes $|A_{z1}|$ larger, recharging rather than draining the battery. That closes the last escape route for the frozen-background caveat: even fully self-consistent, the tachyonic branch does not switch itself off by depletion; the model’s true nonlinear endpoint is the fluid one (§13.1).

Step 13.2.3 (Consequences). Since $\Delta A_{z1} < 0$ where $A_{z1} < 0$, $|\alpha A_{z1}|$ grows: no depletion of the battery. The clean frozen/unfrozen pair shows the global effect is weakly stabilizing (mode amplitude -7% by $|a| \approx 2.5$; annulus compression against the fixed damping boundary), two orders of magnitude too weak to saturate the branch: saturation is by fluid nonlinearity (§13.1), not background drain, and the frozen- A_{z1} approximation is accurate to a few percent everywhere it matters (validated: unfrozen linear rate unchanged, $\gamma = 1.45$ vs eigensolver 1.43–1.46 at the reference blow-up).

14 Warm isothermal fluid closure

14.1 Derivation of the pressure source

Take the first velocity moment of the kinetic equation for beam s with an isotropic thermal spread T (code units, $m = 1$):

$$\partial_t p_i + \partial_j (v_j p_i) = -\partial_i P + F_i, \quad P = nT \quad (\text{isothermal closure}), \quad (14.1)$$

so the only new term relative to the cold model (3.2)–(3.3) is the momentum-density source

$$\boxed{\delta F_i = -T \partial_i n} \quad (14.2)$$

(no extra $1/n$: the sources are momentum-density rates). $T = 0$ recovers the cold system identically. Linearly, the closure adds the sound term $c_s^2 = T$ to the fluid response: the two-stream channel of §12 acquires the standard thermal stabilization when $v_{th} = \sqrt{T} \gtrsim V_0$ (Landau-fluid analogue of $kV < \sqrt{2}\omega_p$ shrinking), which was the intended validation target.

14.2 Why it cannot stabilize the $k_z = 0$ chromo-Weibel

The $k_z = 0$ mode of §2.2 is a filamentation (Weibel-class) instability driven by the momentum-space *anisotropy* of the counter-streaming beams. An isotropic scalar pressure $P = nT$ adds a restoring force only along density gradients of the compressive response; it does not relax the anisotropy that powers filament growth. The scan confirmed this: across $v_{th}/V_0 = 0\text{--}3$ the final $k_z = 0$ amplitude changed by only $\sim 11\%$. (The clean warm-vs-cold validation was subsequently obtained on the KH target point itself: with the corrected control, warm-closure rates match the cold theory at the 1% level, closing T1.4.)

15 Eigenmode seeding theory

15.1 Hermite ground state of the harmonic core

The harmonic core of the given theory, $\partial_x^2 a + (Q_0 - |C_1| x^2) a = 0$ (from (1.2)), is exactly the quantum harmonic oscillator: with $\sigma \equiv |C_1|^{-1/4}$, eigenfunctions and eigenvalues are

$$a_n(x) = H_n\left(\frac{x}{\sigma}\right) e^{-x^2/2\sigma^2}, \quad Q_0 = (2n+1)\sqrt{|C_1|}, \quad (15.1)$$

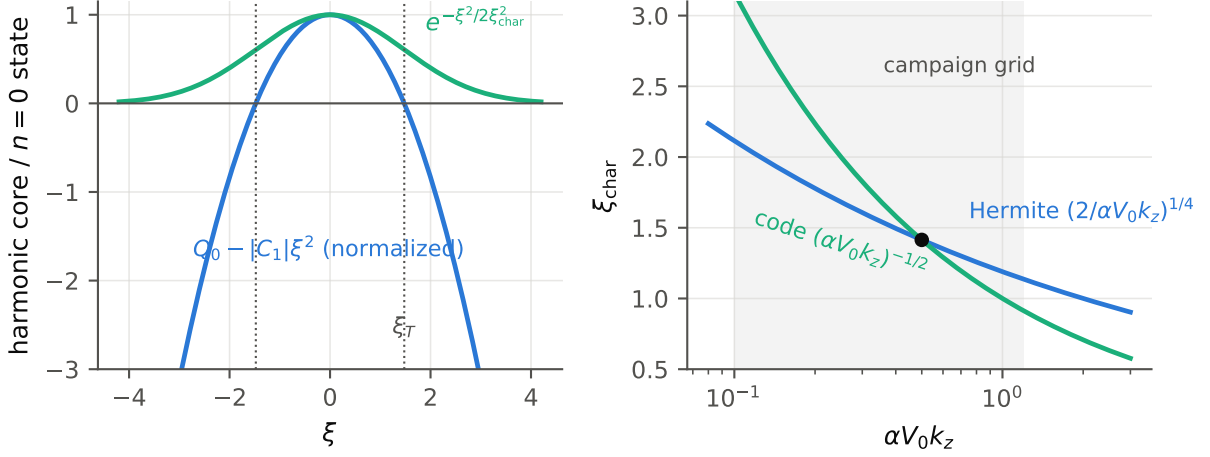


Figure 13: Seeding theory. *Left*: the harmonic core of the given theory with its $n = 0$ Hermite ground state (15.2) ($\alpha = 2$, $V_0 = 0.1$, $k_z = 2$; ξ_T marks the harmonic turning points): eq. (33) of `wkb.pdf` is the oscillator spectrum and the natural seed is this Gaussian. *Right*: the exact Hermite width vs the implemented approximation $(\alpha V_0 k_z)^{-1/2}$: equal at $\alpha V_0 k_z = \frac{1}{2}$, within a few tens of percent over the campaign grid (shaded), but with different power laws outside it.

i.e. eq. (1.4) is the oscillator spectrum, and the $n = 0$ eigenfunction is the Gaussian

$$a_0(x) = \exp\left(-\frac{x^2}{2\xi_{\text{char}}^2}\right), \quad \xi_{\text{char}} = |C_1|^{-1/4} = \left(\frac{2\gamma^2}{\alpha^3 V_0^2}\right)^{1/4} \quad (15.2)$$

(using $|C_1| = \alpha^3 V_0^2 / (2\omega^2)$, $\omega^2 \rightarrow -\gamma^2$ with the magnitude convention of §2.1; x here is the ξ -coordinate of the step-potential form). With the high- k_z quartic balance $\gamma^2 \approx \alpha^2 V_0 / k_z$ (§11),

$$|C_1| \approx \frac{\alpha V_0 k_z}{2} \quad \Rightarrow \quad \xi_{\text{char}} \approx \left(\frac{2}{\alpha V_0 k_z}\right)^{1/4}. \quad (15.3)$$

Remark 15.1 (The implemented width). Campaign 18's init kernel uses the simpler $\xi_{\text{char}}^{\text{code}} = (\alpha k_z V_0)^{-1/2}$, which coincides with (15.3) exactly at $\alpha V_0 k_z = \frac{1}{2}$ (both give $\sqrt{2}$) and tracks it within a few tens of percent over the campaign grid $\alpha V_0 k_z \approx 0.1$ – 1.2 (e.g. $\alpha = 2$, $V_0 = 0.1$, $k_z = 2$: code 1.58 vs Hermite ≈ 1.50); the two have different scaling exponents ($-1/2$ vs $-1/4$), visible in Fig. 13. A width mismatch only launches a transient that decays within a few growth times, so the approximation is operationally adequate; the exact profile is superseded anyway by full six-field seeding (below).

15.2 Why single-field seeding selects wrong modes

An A_z -only (or B_y -only) seed fixes one field's profile and lets the activation chain build the others; the projection of that initial condition onto the eigenmode basis decides what grows first. Concretely (C23): for the target mode at $(\alpha, V_0, k_z) = (1, 0.05, 2)$ the B_y and A_z profiles peak at different ξ (1.64 vs 7.85); an A_z -only Gaussian placed correct-for- A_z deposits the charge response at the wrong location, and a different (co-located, slower) eigenmode wins the race: measured $\gamma = 0.060$ vs target 0.122. The remedy is seeding all six fields with the solver eigenvector (exact amplitude ratios and phases), which removes the transient entirely; conversely §8.5 shows the seed must come from the *dominant* (σ -chased) eigenvector, or the run will faithfully measure an overtone.

15.3 Cascade race condition

Single-field B_y seeding at $k_z \geq 2$ loses a second race: the layer-scale color mixing of §5.2 builds $A_z^{2,3}$ at $\gamma_{\text{casc}} \approx \alpha V_0$ from any residual field, so if $\gamma_{\text{KH}} < \gamma_{\text{casc}}$ the seed decays into the cascade background before the KH chain closes. Gaussian A_z seeding (mode 6) pre-loads the slowest link of the chain $a \rightarrow q_A \rightarrow \text{Lorentz} \rightarrow b$ at the correct width (15.2), which is why it (and its six-field successor) restored clean linear windows at $k_z \geq 2$.

Physical picture. Seeding is choosing the initial overlap. A linear run does not measure “the” growth rate of the system — it measures the growth rate of whatever eigenmode dominates the projection of its initial condition once transients die. Every seeding pathology of the program is this one sentence in different clothes: the C23 stray mode (an A_z -only seed overlapping a co-located slower mode better than the target), the cascade race (a B_y seed whose target projection decays before the chain closes), and the overtone trap of §8.5 (a perfect six-field seed of the *wrong rung*, faithfully grown by a perfect simulation). The cure in every case is the same: know the spectrum first, then seed the intended member of it.

A Helicity projections and the conjugate-extraction identity

For a circular mode the two real fields are

$$A_z^2(x, z) = \text{Re}[\hat{a}(x)e^{ik_z z}], \quad A_z^3(x, z) = \text{Im}[\hat{a}(x)e^{ik_z z}], \quad (\text{A.1})$$

so the complex pair is single-signed in k_z : $a = A_z^2 + iA_z^3 = \hat{a}(x)e^{ik_z z}$ exactly. Its z -DFT therefore has *all* its weight at $+k_z$ and none at $-k_z$:

$$\tilde{a}(+k_z) = \hat{a}, \quad \tilde{a}(-k_z) = 0. \quad (\text{A.2})$$

Writing the projections in real arithmetic with $c = \cos k_z z$, $s = \sin k_z z$:

$$\tilde{a}(\pm k_z) \propto \sum_z [(A_z^2 c \pm A_z^3 s) + i(\mp A_z^2 s + A_z^3 c)]. \quad (\text{A.3})$$

The extraction bug of 2026-07-04→07-17 computed the lower sign (the $-k_z$, conjugate-helicity projection): by (A.2) this is *identically zero* for a pure circular mode, and what it actually returned on real data was the small opposite-helicity contamination — hence amplitudes at the e^{-25} floor and the triage rule “first-amplitude $< e^{-25} \Rightarrow$ invalid extraction”. The mirror symmetry of §4.3 guarantees the physical information lives in one helicity per side of the layer, so the corrected single-sign projection loses nothing.

B Filters, sponges, and hyperdiffusion as linear operators

DFT band filter. `kernel_ym_subtract_kz_range` applies, each step, the projection $f \mapsto f - \sum_{k \in \mathcal{B}} \Pi_k f$, where Π_k is the exact DFT projector onto z -mode k and $\mathcal{B} = \{1, \dots, k_{\text{lo}}\} \cup \{k_{\text{mode}} + 1, \dots, k_{\text{hi}}\}$: an idempotent linear operator that removes the two-stream band (12.6) while leaving the measured mode untouched. The same projector acts on the fluid p_z ’s. (Empirical constraint, recorded not derived: keep $k_{\text{hi}} \leq N_Z/4.5$ so the projector’s stencil does not interact with the FCT limiter’s near-Nyquist deposits.)

Sponge. `kernel_ym_sponge` adds $\partial_t f = -\sigma_s(\xi)f$ with $\sigma_s = \sigma(|\xi| - \xi_{\text{sponge}})_+$: a non-Hermitian absorbing potential. In the eigenproblem it appears as $-\sigma_s(\xi)$ on the diagonal of all blocks; it removes the outer branch of §5.2 when $\xi_{\text{sponge}} < \xi_{\text{crit}}$ (legitimate, by the battery argument) at the cost of the few-percent well compression measured in the sponge-vs-cut comparisons.

Hard cut. `xi_cut` zeroes all wave fields outside ξ_w : Dirichlet/Neumann wall pair of §6.4.

Hyperdiffusion. The z -hyperdiffusion adds $\partial_t f = -\mu \partial_z^4 f$, i.e. damping μk^4 per mode: with $\mu = 5 \times 10^{-5}$ it removes grid-scale noise while its drag on a retained mode k is $\mu k^4 \ll \gamma$ for all production k (e.g. $k = 6$: $\mu k^4 = 0.065\gamma$ at $\gamma = 0.1$; the design condition is $1/(\mu k^4) \gg 1/\gamma$).

C Validation ledger

Derived result	Numerical anchor
Quartic $k_z=0$ chromo-Weibel rate (2.8)	sim vs formula: 0.5%; $\gamma = 0.284$ reproduced at the $\alpha = 1$, $V_0 = 0.05$ reference
Six-field system (4.27)	eigensolver vs 2D CUDA growth rates: 1–4% (plateau-confirmed clean cores: median 6–10% archive-wide)
Exact reduction (6.5)/(6.12)	reduced operator vs 6-field ARPACK: rel. diff. 0.0 ($< 10^{-10}$) at 7 points, $\alpha = 1-3$, $V_0 = 0.03-0.1$, $k_z = 1-7$
Tachyonic law (5.4)–(5.5)	analytic envelope = float64 eigensolver = float32 CUDA rates at all outer eigenmode peaks; blow-up $\gamma = 1.45$ vs 1.43–1.46
Ceiling (7.5)	dominant-branch peaks at 95–99% of $(\alpha V_0^2)^{1/3}$ in every series; T2.5 collapse $\gamma_{\text{peak}}/\Gamma = 0.977 \pm 0.011$
Model B quantization (§8.4)	$\leq 2\%$ (median $\leq 0.2\%$) vs σ -chased $n=0$ eigenvalues, 12 series, $k_z \geq 1.5$
Model A closed form (8.11)	$\sim 3\%$ near/above peak; peak located to ± 0.5 in k_z
Peak formula (9.2)	± 0.6 in k_z across 11 series; window test: cut 11 $\rightarrow$ 25 moved peak 4.5 $\rightarrow$ 5.5 (formula 4.45 $\rightarrow$ 5.85)
Two-stream band (12.6)	$k_{\text{ts}} \approx 14$ at $V_0 = 0.1$ (documented band edge)
Back-reaction law (13.3)	$\Delta A_{z1} = -0.04 a ^2$ over four decades; endpoint -0.242 vs -0.242
Warm closure (§14)	warm vs cold theory at the KH target: 1% (corrected control)